

# Mahler measures at interior CM points: proofs of two conjectures of Samart

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## Declaration on the use of AI tools

The research reported in this article — including the computational exploration, the discovery of the proof strategy, the machine-certified verifications, and the preparation of the manuscript — was carried out by the author with the assistance of the AI system *Kimi* (Moonshot AI). All mathematical content, including every proof and every certified computation, has been checked and verified by the author, who takes full responsibility for the correctness and integrity of the article. All certification scripts are available for independent verification (see Section 6).

## Abstract

We prove two conjectures of Samart as identities of genuine Mahler measures. The first is the case  $k = 1$  of his evaluations for the family  $(x + x^{-1})(y + y^{-1})(z + z^{-1}) + k^{1/2}$ :

$$m((x + x^{-1})(y + y^{-1})(z + z^{-1}) + 1) = 4L'(g_7, 0),$$

where  $g_7(\tau) = \eta(\tau)^3 \eta(7\tau)^3$  is the unique newform of  $S_3(\Gamma_0(7), \chi_{-7})$  (LMFDB label 7.3.b.a). The second is the conjugate pair of quadratic CM entries of his 2015 table:

$$n_2\left(\frac{47 \pm 45\sqrt{-7}}{2}\right) = \frac{4}{7}\left(54L'(g_7, 0) + L'(\chi_{-7}, -1)\right),$$

where  $n_2(s) := 2m((x + x^{-1})(y + y^{-1})(z + z^{-1}) + \sqrt{s})$ . In both cases the parameter lies inside (or on the boundary of) the critical locus, where the identity  $m = \operatorname{Re} \tilde{m}$  with the holomorphic (modified) Mahler measure is not guaranteed: the  $L$ -value side was computed by Samart (conditionally) and, at the  $\operatorname{Re} \tilde{m}$  level, by Fei, but the passage to the true Mahler measure at these points was open. For the first theorem we close the gap by a differential-comparison continuation: the parameter derivative of the Mahler measure is a single-valued period integral, the holomorphic differential identity spreads from Samart's region by the identity theorem, and an explicit certified path together with the continuity of Mahler measure finishes the proof at the interior point. For the second theorem no continuation is needed: the second preimage of the parameter under the modular parametrization is the Fricke partner of Samart's CM point and lies inside Samart's proved region, so his formula applies directly; the evaluation is then an exact CM lattice-sum computation with an imprimitive Hecke-sum factor. All non-elementary inputs are either quoted theorems or interval-arithmetic certificates, and all identities are confirmed numerically to 41–60 digits.

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# 1 Introduction

## 1.1 Statement of the results

Let

$$f(x, y, z) := (x + x^{-1})(y + y^{-1})(z + z^{-1}),$$

and denote by  $m(P)$  the logarithmic Mahler measure of a Laurent polynomial,

$$m(P) = \int_{\mathbb{T}^3} \log |P(x, y, z)| \frac{dx}{2\pi i x} \frac{dy}{2\pi i y} \frac{dz}{2\pi i z}, \quad \mathbb{T} = \{|w| = 1\}.$$

Following Samart [1, 3] we write

$$n_2(s) := 2m(f + \sqrt{s}), \quad s \in \mathbb{C}, \quad (1)$$

where the value does not depend on the choice of the square root, since  $f(-x, y, z) = -f(x, y, z)$  and Mahler measure is invariant under  $x \mapsto -x$ ; hence  $m(f + c) = m(f - c)$  for every  $c \in \mathbb{C}$ . (In [1] the same function is denoted  $f_2$ .) We prove:

**Theorem 1.1** (Theorem A; Samart's conjecture for  $k = 1$ , [1, 3]).

$$m(f + 1) = 4L'(g_7, 0), \quad \text{equivalently} \quad n_2(1) = 8L'(g_7, 0),$$

where

$$g_7(\tau) = \eta(\tau)^3 \eta(7\tau)^3 = \sum_{n \geq 1} a_n q^n = q - 3q^2 + 5q^4 - 7q^7 - 3q^8 + 9q^9 + \dots, \quad q = e^{2\pi i \tau},$$

is the unique normalized newform of  $S_3(\Gamma_0(7), \chi_{-7})$  (LMFDB label 7.3.b.a [14]).

**Theorem 1.2** (Theorem B; Samart's Table- $n_2$  conjugate pair, [3]).

$$n_2\left(\frac{47 \pm 45\sqrt{-7}}{2}\right) = \frac{4}{7} \left(54L'(g_7, 0) + L'(\chi_{-7}, -1)\right),$$

where  $\chi_{-7}$  is the odd quadratic Dirichlet character of conductor 7. The two conjugate identities hold simultaneously (and are equal, since  $n_2(\bar{s}) = n_2(s)$ ).

Numerically, the true Mahler measures computed by direct torus integration and the  $L$ -value expressions computed from the functional equations agree to 41 and 50 digits respectively:

$$\begin{aligned} m(f + 1) &= 0.4106864311156080448418263795931716559956\dots, \\ 4L'(g_7, 0) &= 0.4106864311156080448418263795931716559956\dots \quad (\text{difference } 6 \times 10^{-42}), \\ n_2\left(\frac{47 \pm 45\sqrt{-7}}{2}\right) &= 4.13826815832141945836337668037433080930698376\dots \\ &= \frac{4}{7} (54L'(g_7, 0) + L'(\chi_{-7}, -1)); \end{aligned}$$

the agreement in the second line is  $10^{-50}$ ; see Section 7.

## 1.2 History and the interior-point problem

In [1] Samart expressed  $n_2(s)$ , for  $s$  in the image of a modular parametrization  $s_2(\tau)$  (see §2.2), as an Eisenstein–Kronecker series valid for  $\text{Im } \tau \geq 1/2$ , deduced the exterior evaluations  $n_2(64)$  and  $n_2(256)$ , and conjectured on numerical evidence a number of further  $L$ -value evaluations, among them  $n_2(1) = 8L'(g_7, 0)$ . In [3] he organized the conjectures around CM points of the parametrization: his tables attach to each CM value of  $s$  a conjectural expression of  $n_2(s)$  as a rational linear combination of  $L$ -values of CM weight-three newforms and Dirichlet characters. The rows  $s = 1$  (attached to  $\tau_0 = (3 + \sqrt{-7})/8$ ) and  $s = (47 \pm 45\sqrt{-7})/2$  (attached to  $\tau' = (\pm 1 + \sqrt{-7})/8$ ) of that table are exactly Theorems A and B.

Several entries of Samart’s lists have been proved:  $k = 16$  (an interior parameter, by regulator methods) and  $k = -104 \pm 60\sqrt{3}$ ,  $4096$ ,  $-2024 \pm 765\sqrt{7}$  by Zheng–Guo–Qin [5]; further cases including negative real parameters by Guo–Peng–Qin [6]; and the conjectures for the companion family  $f_3$  by He–Ye [7]. Fei [4] treated 23 families of three-variable Laurent polynomials (3D Landau–Ginzburg potentials) uniformly and proved, for all rational singular moduli — including the point  $s = 1$  of Theorem A — identities expressing the *real part of the holomorphic (modified) Mahler measure*  $\text{Re } \tilde{m}$  in terms of  $L$ -values. However, at parameters inside the critical locus the identity

$$m(f - c) \stackrel{?}{=} \text{Re } \tilde{m}(c)$$

is not guaranteed: the holomorphic Mahler measure  $\tilde{m}$  is a multivalued analytic object whose real part agrees with  $m(f - c)$  only off the critical segment, a point stressed already by Rodríguez Villegas [9] and recalled by Fei himself. Consequently the genuine Mahler measure identities at interior CM points remained open even where the  $\text{Re } \tilde{m}$ -identities were known. Theorem A is, to our knowledge, the first complete proof of a genuine Mahler measure identity at the interior point  $s = 1$ . Theorem B concerns non-real (quadratic) CM parameters, which are not covered by Fei’s rational-singular-modulus results, and is entirely new.

The structural difficulty can be seen on the modular side. Samart’s exponential-sum formula is proved only for  $\text{Im } \tau \geq 1/2$ , where  $s_2(\tau)$  stays away from the critical locus. The point  $\tau_0$  with  $s_2(\tau_0) = 1$  has  $\text{Im } \tau_0 = \sqrt{7}/8 \approx 0.331 < 1/2$ , and the value  $s = 1$  lies *inside* the critical interval: the real loci of  $s_2$  in a fundamental domain reach only  $k \leq 0$  and  $k \geq 64$ , so  $k = 1$  is an isolated interior point with no real-analyticity bridge. Likewise  $\tau' = (1 + \sqrt{-7})/8$  has  $\text{Im } \tau' = \sqrt{7}/8 < 1/2$ , and evaluating Samart’s series at  $\tau'$  produces a *wrong-sheet* value (Remark 5.12); Theorem B circumvents this by the Fricke trick described below.

## 1.3 Methods

**Theorem A: differential-comparison continuation.** Instead of tracking branches of the holomorphic Mahler measure we compare *differentials*. The parameter derivative of the Mahler measure is a single-valued period integral  $\Omega$  on the complement of the critical locus (Lemma 2.12), so differentiating Samart’s identity yields a holomorphic differential identity  $\Psi \equiv 0$  that spreads from Samart’s region to a whole connected component  $W$  of the good domain by the identity theorem (Proposition 3.4). An abstract propagation principle (Lemma 3.1) then reduces the conclusion to one topological input — that the target point  $\tau_0$  lies in  $\overline{W}$  — supplied by an explicit certified path (Proposition 4.9), plus the continuity of Mahler measure at the endpoint. The remaining pillar is an exact CM evaluation of Samart’s series at  $\tau_0$  (P1, §4.1): the lattice-sum form of the series involves the ring of integers of  $\mathbb{Q}(\sqrt{-7})$  and a principal ideal, and the parasitic  $\zeta_K(2)$ -terms cancel exactly.

**Theorem B: the Fricke trick.** The parametrization  $s_2$  has degree 2 on  $X_0(4)$ , and the two preimages of  $s = (47 + 45\sqrt{-7})/2$  relevant here are Samart's point  $\tau'$  (outside the proved region) and its Fricke partner  $\tau_w = -1/(4\tau') = (-1 + \sqrt{-7})/4$ , which satisfies  $\text{Im } \tau_w = \sqrt{7}/4 \geq 1/2$  and hence lies *inside* Samart's proved region. Since  $s_2$  is exactly invariant under the Fricke involution (Lemma 5.1), Samart's theorem applies at  $\tau_w$  directly and no continuation is needed. The evaluation of Samart's series at  $\tau_w$  (P1, §5.3) is again a lattice-sum computation, but now one lattice is the non-maximal order of discriminant  $-28$ : the sums are imprimitive Hecke sums with an exact Euler factor  $23/16$ , the  $\zeta_K(2)$ -terms do *not* cancel, and their survival is precisely the source of the Dirichlet term  $L'(\chi_{-7}, -1)$ .

All non-elementary inputs are either quoted theorems or interval-arithmetic certificates (Section 6); every identity is also confirmed numerically to 41–60 digits (Section 7).

## 1.4 Notation

Throughout,  $\tau = x + iy \in \mathbb{H}$ ,  $q = e^{2\pi i \tau}$ ,  $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ ,  $\Delta = \eta^{24}$ ,  $\mathbb{T}$  is the unit circle, and  $\sum'$  denotes omission of the zero term. We write

$$M_7 := L'(g_7, 0), \quad d_7 := L'(\chi_{-7}, -1),$$

matching Samart's shorthand in [3]. The field  $K = \mathbb{Q}(\sqrt{-7})$  has ring of integers  $\mathcal{O}_K = \mathbb{Z}[\omega]$ ,  $\omega = (1 + \sqrt{-7})/2$ , class number  $h(-7) = 1$  and units  $\mathcal{O}_K^\times = \{\pm 1\}$ ; we also use  $\pi := (1 + \sqrt{-7})/2$ , satisfying  $\pi^2 - \pi + 2 = 0$ ,

$$\pi + \bar{\pi} = 1, \quad \pi \bar{\pi} = 2 \quad (\text{so } (2) = (\pi)(\bar{\pi}) \text{ splits}), \quad \pi^2 + \bar{\pi}^2 = -3, \quad \text{Re } \pi^2 = -\frac{3}{2}. \quad (2)$$

The Jacobi theta constants are taken with  $q_\vartheta = e^{\pi i \tau}$ :

$$\begin{aligned} \vartheta_2(\tau) &= 2q_\vartheta^{1/4} \prod_{n \geq 1} (1 - q_\vartheta^{2n})(1 + q_\vartheta^{2n}), & \vartheta_3(\tau) &= \prod_{n \geq 1} (1 - q_\vartheta^{2n})(1 + q_\vartheta^{2n-1}), \\ \vartheta_4(\tau) &= \prod_{n \geq 1} (1 - q_\vartheta^{2n})(1 - q_\vartheta^{2n-1}), & \lambda(\tau) &= (\vartheta_2(\tau)/\vartheta_3(\tau))^4. \end{aligned}$$

## 2 Preliminaries

### 2.1 Mahler measure: continuity and the holomorphic Mahler measure

**Lemma 2.1** (Continuity of Mahler measure in coefficients). *The function  $c \mapsto m(f - c)$  is continuous on  $\mathbb{C}$ . Consequently  $n_2(s) = 2m(f + \sqrt{s})$  is a well-defined continuous function of  $s \in \mathbb{C}$ , independent of the choice of  $\sqrt{s}$ .*

*Proof.* Write  $f(x, y, z) = C(z + z^{-1})$  with  $C = (x + x^{-1})(y + y^{-1})$ . Mahler measure is invariant under multiplication by the monomial  $z$ , so

$$m(f - c) = \int_{\mathbb{T}^2} M(Cz^2 - cz + C) d\mu(x, y),$$

where  $M$  denotes the one-variable Mahler measure and  $\mu$  Haar measure. For  $az^2 + bz + a$ ,  $M = \log |a| + \sum_i \log^+ |\alpha_i|$  over the roots  $\alpha_i$ , which is a continuous function of the coefficient vector  $(a, b)$ : the roots vary continuously and  $\log^+ |\cdot|$  is continuous. Hence for every  $(x, y) \in \mathbb{T}^2$  with  $C \neq 0$  the integrand is continuous in  $c$ ; the exceptional set  $\{C = 0\}$  has measure zero. Moreover

$$M(Cz^2 - cz + C) \leq \log(2|C| + |c| + 2) + \log 2,$$

and since  $|\alpha_1\alpha_2| = 1$  also  $M(Cz^2 - cz + C) \geq \log |C|$ ; the upper bound is nonnegative, so  $|M| \leq |\log |C|| + \log(2|C| + |c| + 2) + \log 2$ . Both terms are integrable over  $\mathbb{T}^2$  uniformly for  $c$  in a compact set ( $\log |C|$  is the Mahler integrand of the nonzero Laurent polynomial  $C$ ). Dominated convergence gives the claim. For  $n_2$ : the two branches of  $\sqrt{s}$  differ by a sign, and  $m(f + c) = m(f - c)$  by the substitution  $x \mapsto -x$ , so  $n_2$  is well defined; its continuity follows from the continuity of  $c \mapsto m(f - c)$  and of  $\sqrt{\cdot}$ .  $\square$

The range  $f(\mathbb{T}^3)$  is the segment  $K = [-8, 8]$ , the *critical segment* of the family  $f - c$ . The following is a construction of Villegas [9]:

**Lemma 2.2** (Holomorphic Mahler measure). *There is a holomorphic function  $\tilde{m}$  on the universal cover of  $\mathbb{C} \setminus K$ , whose imaginary part is single-valued up to additive real constants, such that*

$$m(f - c) = \operatorname{Re} \tilde{m}(c) \quad (c \in \mathbb{C} \setminus K).$$

In particular  $c \mapsto m(f - c)$  is real-analytic on  $\mathbb{C} \setminus K$ .

Rogers [8] gave an explicit hypergeometric representative, valid for  $|k| \geq 64$  with  $k \neq 64$ :

$$n_2(k) = \operatorname{Re} \left[ \log k - \frac{8}{k} {}_5F_4 \left( \frac{3}{2}, \frac{3}{2}, \frac{3}{2}, 1, 1; 2, 2, 2, 2; \frac{64}{k} \right) \right], \quad (3)$$

the  ${}_5F_4$  series converging for  $|64/k| \leq 1$ , including the boundary circle by  $\sum b - \sum a = 3/2 > 0$ . We use (3) only for numerical cross-checks (at the boundary points  $k_{\pm}$ ,  $|k_{\pm}| = 64$ ).

*Remark 2.3* (The two cuts). As a function of the shift  $c$ , the bad locus is the segment  $K = [-8, 8]$ ; as a function of Samart's parameter  $k$  with  $c = \sqrt{k}$ , the bad locus is the segment  $[0, 64]$ , since  $c^2 \in [0, 64]$  with  $c \notin [-8, 8]$  is impossible. The continuation machine of Section 3 is formulated in the  $k$ -variable with the cut  $[0, 64]$ . This distinction is invisible at  $k = 1$  (where  $\sqrt{k} = 1$ ), but it is essential for the correct formulation of the propagation and for understanding the wrong-sheet phenomenon of Remark 5.12.

## 2.2 The modular parametrization $s_2$

Define

$$s_2(\tau) := -\frac{\Delta(\tau + \frac{1}{2})}{\Delta(2\tau + 1)}. \quad (4)$$

**Lemma 2.4** (Product expansion).

$$s_2(\tau) = q^{-1} \prod_{n \geq 1} (1 + q^{2n-1})^{24} = q^{-1} + 24 + 276q + 2048q^2 + 11202q^3 + 49152q^4 + 184024q^5 + \dots \quad (5)$$

In particular  $s_2$  is holomorphic on  $\mathbb{H}$ , has integral  $q$ -expansion, and satisfies  $s_2(\tau + 1) = s_2(\tau)$ .

*Proof.* From  $\eta(\tau + \frac{1}{2}) = e^{\pi i(\tau + 1/2)/12} \prod_{n \geq 1} (1 - (-1)^n q^n)$  we get  $\Delta(\tau + \frac{1}{2}) = e^{2\pi i(\tau + 1/2)} \prod (1 - (-1)^n q^n)^{24} = -q \prod (1 - (-1)^n q^n)^{24}$ , and from  $\eta(2\tau + 1) = e^{\pi i(2\tau + 1)/12} \prod (1 - q^{2n})$  we get  $\Delta(2\tau + 1) = q^2 \prod (1 - q^{2n})^{24}$ . Splitting  $\prod (1 - (-1)^n q^n) = \prod (1 + q^{2n-1}) \prod (1 - q^{2n})$  and cancelling  $\prod (1 - q^{2n})$  gives (5); the expansion is a finite integer computation. Periodicity follows from  $\Delta(\tau + \frac{3}{2}) = \Delta(\tau + \frac{1}{2})$  and  $\Delta(2\tau + 3) = \Delta(2\tau + 1)$ .  $\square$

**Theorem 2.5** (The  $\lambda$ -identity for  $s_2$ ). *For every  $\tau \in \mathbb{H}$ ,*

$$s_2(\tau) = \frac{16}{\lambda(2\tau)(1 - \lambda(2\tau))}. \quad (6)$$

*Proof.* By Jacobi's identity  $\vartheta_3^4 = \vartheta_2^4 + \vartheta_4^4$ ,  $\lambda = \vartheta_2^4/\vartheta_3^4$  and  $1 - \lambda = \vartheta_4^4/\vartheta_3^4$ , so  $16/(\lambda(1 - \lambda)) = 16\vartheta_3^8/(\vartheta_2^4\vartheta_4^4)$ , and

$$\frac{16\vartheta_3(2\tau)^8}{\vartheta_2(2\tau)^4\vartheta_4(2\tau)^4} = q^{-1} \prod_{n \geq 1} \frac{(1 + q^{2n-1})^{16}}{(1 + q^{2n})^8(1 - q^{2n-1})^8}.$$

Now  $(1 + q^{2n-1})^{16}(1 - q^{2n-1})^{-8} = (1 + q^{2n-1})^{24}[(1 + q^{2n-1})(1 - q^{2n-1})]^{-8} = (1 + q^{2n-1})^{24}(1 - q^{4n-2})^{-8}$ , hence

$$\frac{16\vartheta_3(2\tau)^8}{\vartheta_2(2\tau)^4\vartheta_4(2\tau)^4} = q^{-1} \prod_{n \geq 1} (1 + q^{2n-1})^{24} \cdot \left[ \prod_{n \geq 1} (1 + q^{2n})(1 - q^{4n-2}) \right]^{-8}.$$

The second product equals 1 by Euler's identity

$$\prod_{n \geq 1} (1 + q^{2n}) = \frac{\prod_{n \geq 1} (1 - q^{4n})}{\prod_{n \geq 1} (1 - q^{2n})} = \frac{\prod_{n \geq 1} (1 - q^{4n})}{\prod_{n \geq 1} (1 - q^{4n-2}) \prod_{n \geq 1} (1 - q^{4n})} = \prod_{n \geq 1} \frac{1}{1 - q^{4n-2}}.$$

Therefore the right side of (6) equals  $q^{-1} \prod (1 + q^{2n-1})^{24} = s_2(\tau)$  by Lemma 2.4. (An exact integer series check of the identity to order  $q^{37}$  is included in `cert0_s2_eq_1.py` and `cert0_s2_n2pair.py` as a sanity check.)  $\square$

*Remark 2.6.* The function  $s_2$  is  $\Gamma_0(4)$ -invariant (since  $\lambda(2\tau)$  is) and has degree 2 on  $X_0(4)$ : by (5) it has a simple pole at the cusp  $\infty$ ; as  $t \rightarrow 0^+$ ,  $\prod (1 + e^{-2\pi t(2n-1)})^{24} \rightarrow e^{\pi/(2t)}$  super-exponentially, so  $s_2(it) \rightarrow \infty$  at the cusp 0 as well, while  $s_2(\frac{1}{2} + it) \rightarrow 0$  at the cusp  $\frac{1}{2}$ . It is therefore *not* a Hauptmodul; the two preimages of a generic value are related by the Fricke involution  $\tau \mapsto -1/(4\tau)$  (Lemma 5.1).

### 2.3 The newform $g_7$ and the constants $M_7, d_7$

The eta product  $g_7 = \eta(\tau)^3 \eta(7\tau)^3$  lies in  $S_3(\Gamma_0(7), \chi_{-7})$ . Its coefficients are given by the Jacobi triple product:

$$a_n = \sum_{\substack{i, j \geq 0 \\ T_i + 7T_j = n-1}} (-1)^{i+j} (2i+1)(2j+1), \quad T_i = \frac{i(i+1)}{2}, \quad (7)$$

since Jacobi's identity  $\eta(\tau)^3 = \sum_{i \geq 0} (-1)^i (2i+1) q^{T_i+1/8}$  gives  $g_7 = \sum_{i, j \geq 0} (-1)^{i+j} (2i+1)(2j+1) q^{T_i+7T_j+1}$ , and collecting  $T_i + 7T_j + 1 = n$  gives (7).

**Proposition 2.7** (Theta identity; proved in Appendix B).

$$g_7(\tau) = \frac{1}{2} \sum'_{\alpha \in \mathcal{O}_K} \alpha^2 q^{N(\alpha)}. \quad (8)$$

Consequently, for  $\text{Re } s > 2$ ,

$$\sum'_{\alpha \in \mathcal{O}_K} \frac{\alpha^2}{|\alpha|^{2s}} = 2L(g_7, s), \quad \sum'_{\alpha \in \mathcal{O}_K} \frac{\bar{\alpha}^2}{|\alpha|^{2s}} = 2L(g_7, s), \quad (9)$$

the second equality following from the first by complex conjugation since  $L(g_7, s)$  has rational coefficients, and

$$\sum'_{\alpha \in \mathcal{O}_K} |\alpha|^{-4} = 2\zeta_K(2). \quad (10)$$

**Lemma 2.8** (Functional equations). (i) *The completed  $L$ -function  $\Lambda(s) := 7^{s/2}(2\pi)^{-s}\Gamma(s)L(g_7, s)$  satisfies  $\Lambda(s) = \Lambda(3-s)$  (root number  $+1$ ). Hence  $L(g_7, 0) = 0$  and*

$$M_7 = L'(g_7, 0) = \Lambda(3) = 7^{3/2}(2\pi)^{-3}\Gamma(3)L(g_7, 3) = \frac{7\sqrt{7}}{4\pi^3}L(g_7, 3). \quad (11)$$

(ii) *For  $\chi = \chi_{-7}$ , the completed  $L$ -function  $\Lambda(s, \chi) := (7/\pi)^{(s+1)/2}\Gamma((s+1)/2)L(s, \chi)$  satisfies  $\Lambda(s, \chi) = \Lambda(1-s, \chi)$  (root number  $+1$ ). Since  $\chi$  is odd,  $L(\chi, -1) = 0$ , and the simple pole of  $\Gamma((s+1)/2)$  at  $s = -1$  (residue 2 in  $s$ ) against the simple zero of  $L$  gives  $\Lambda(-1, \chi) = 2d_7$ ; hence*

$$2d_7 = \Lambda(2, \chi) = \left(\frac{7}{\pi}\right)^{3/2}\Gamma\left(\frac{3}{2}\right)L(\chi_{-7}, 2), \quad \text{i.e.} \quad d_7 = \frac{7\sqrt{7}}{4\pi}L(\chi_{-7}, 2). \quad (12)$$

(iii) *Since  $\zeta_K(2) = \zeta(2)L(\chi_{-7}, 2)$ ,*

$$\zeta_K(2) = \frac{\pi^2}{6}L(\chi_{-7}, 2) = \frac{2\pi^3}{21\sqrt{7}}d_7. \quad (13)$$

*Proof.* The functional equation in (i) is standard for the CM form  $g_7$  (Proposition 2.7); the root number  $+1$  is known for this newform and was independently verified numerically to 52 digits by comparing the two sides of the functional equation (`lvalue_g7.py`). Since  $\Gamma$  has a simple pole at  $s = 0$  while  $\Lambda(0) = \Lambda(3)$  is finite,  $L(g_7, s)$  vanishes at  $s = 0$ , and  $L'(g_7, 0) = \lim_{s \rightarrow 0} \Gamma(s)L(g_7, s) = \Lambda(3)$ ; finally  $\Lambda(3) = 7^{3/2}(2\pi)^{-3} \cdot 2 \cdot L(g_7, 3)$  as stated. Part (ii) is the classical functional equation of an odd primitive Dirichlet character (the Gauss sum of  $\chi_{-7}$  is  $i\sqrt{7}$ ), and (iii) is the factorization  $\zeta_K = \zeta \cdot L(\chi_{-7}, \cdot)$  at  $s = 2$  combined with (12). Numerical confirmations are collected in Section 7.  $\square$

## 2.4 Samart's exponential-sum formula and its lattice form

For  $j \geq 1$  and  $\tau \in \mathbb{H}$  define

$$U_j(\tau) := \sum_{m \neq 0} \frac{1}{m} \sum_{n \in \mathbb{Z}} (jm\tau + n)^{-3} = 2\pi^3 \sum_{m \geq 1} \frac{\cos(j\pi m\tau)}{m \sin^3(j\pi m\tau)}, \quad (14)$$

where the second form follows by pairing the indices  $m$  and  $-m$  and applying the partial fraction identity  $\sum_{n \in \mathbb{Z}} (w + n)^{-3} = \pi^3 \cos(\pi w) / \sin^3(\pi w)$  to  $w = jm\tau$ ; the double series is absolutely convergent since  $\sum_{m \neq 0} |m|^{-1} \sum_n |jm\tau + n|^{-3}$  is  $O(\sum_m |m|^{-3} y^{-2}) < \infty$ , locally uniformly in  $\tau$ . Define Samart's exponential-sum expression

$$\text{EK}(\tau) := \text{Im} \left[ 2\pi\tau + \frac{2}{\pi^3} (U_1(\tau) - 4U_4(\tau)) \right], \quad (15)$$

real-analytic on all of  $\mathbb{H}$ .

**Theorem 2.9** (Samart, [1, Prop. 2.1(i)]). *For  $\tau \in \mathbb{H}$  with  $\text{Im } \tau \geq 1/2$  one has*

$$n_2(s_2(\tau)) = \text{EK}(\tau), \quad (16)$$

*i.e.  $2\text{m}(f + \sqrt{s_2(\tau)}) = \text{EK}(\tau)$  for either choice of the square root. In particular, on the imaginary axis  $s_2(it) \geq 64$  for  $t \geq 1/2$  and  $n_2(s_2(it)) = \text{EK}(it)$ .*

Samart's proof combines Lemma 2.2 with an explicit computation of  $\tilde{m} \circ s_2$  via the Picard–Fuchs equation of the family and a Wronskian argument; we take it as a quoted theorem. Its holomorphic content is the identity

$$\tilde{m}(\sqrt{s_2(\tau)}) = E(\tau), \quad E(\tau) := -\pi i \tau - \frac{i}{\pi^3} (U_1(\tau) - 4U_4(\tau)), \quad (17)$$

on the region where his series converge, and  $EK = 2 \operatorname{Re} E$ .

For the CM evaluations we use the lattice-sum form of EK (which is in fact the form stated in [1, Prop. 2.1]). For  $\tau = x + iy \in \mathbb{H}$  and  $d \geq 1$  define the double series

$$T_d(\tau) := \sum_{(m,n) \in \mathbb{Z}^2} ' \left[ \frac{4(dm\tau + n)^2}{|dm\tau + n|^6} - \frac{1}{|dm\tau + n|^4} \right], \quad (18)$$

summed by rows in  $m$  (each row is absolutely convergent and the row sums decay like  $|m|^{-2}$ , as Poisson summation shows; the  $m = 0$  row is  $\sum_{n \neq 0} (4n^2 \cdot n^{-6} - n^{-4}) = 6\zeta(4) = \pi^4/15$ ).

**Lemma 2.10** (Key identity). *For every  $\tau \in \mathbb{H}$ ,*

$$EK(\tau) = \frac{2 \operatorname{Im} \tau}{\pi^3} \left( -T_1(\tau) + 16T_4(\tau) \right). \quad (19)$$

*Proof.* Write  $w = jm\tau + n = u + iv$  with  $u = jmx + n$ ,  $v = jmy$ . From  $w^{-3} - \bar{w}^{-3} = (\bar{w}^3 - w^3)/|w|^6 = -2i(3u^2v - v^3)/|w|^6$  and  $3u^2 - v^2 = 4u^2 - |w|^2$ ,

$$\begin{aligned} 2i \operatorname{Im} U_j &= U_j - \overline{U_j} = \sum_{m \neq 0} \frac{1}{m} \sum_{n \in \mathbb{Z}} [(jm\tau + n)^{-3} - (jm\bar{\tau} + n)^{-3}] \\ &= \sum_{m \neq 0} \frac{1}{m} \sum_{n \in \mathbb{Z}} [-2iv(3u^2 - v^2)] |w|^{-6} = -2i jy \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \left[ \frac{4u^2}{|w|^6} - \frac{1}{|w|^4} \right], \end{aligned}$$

where we used  $(1/m)v = jy$  (note  $v = jmy$  has the sign of  $m$ , so the factor  $1/m$  cancels it exactly). The inner double sum is  $T_j(\tau)$  minus its  $m = 0$  row  $\pi^4/15$ ; hence

$$\operatorname{Im} U_j = -jy \left( T_j(\tau) - \frac{\pi^4}{15} \right).$$

Substituting into (15),

$$EK(\tau) = 2\pi y + \frac{2}{\pi^3} \left[ -y(T_1 - \frac{\pi^4}{15}) + 16y(T_4 - \frac{\pi^4}{15}) \right] = 2\pi y + \frac{2y}{\pi^3} (-T_1 + 16T_4) - \frac{2y}{\pi^3} \cdot \pi^4,$$

and the last term is  $2\pi y$ , so they cancel, giving (19). (The  $m = 0$  rows of  $T_1$  and  $T_4$  contribute  $(-1 + 16)\pi^4/15 = \pi^4$  to the combination, which is exactly the polynomial term  $2\pi\tau$  of Samart's formula.)  $\square$

*Remark 2.11* (Alternative form). Since  $4u^2 = (w + \bar{w})^2 = w^2 + 2|w|^2 + \bar{w}^2$ ,

$$T_d(\tau) = \sum_{\lambda \in \Lambda_d(\tau)} ' \left[ 2 \operatorname{Re} \frac{\bar{\lambda}^2}{|\lambda|^6} + \frac{1}{|\lambda|^4} \right], \quad \Lambda_d(\tau) := \mathbb{Z} + \mathbb{Z} d\tau, \quad (20)$$

which is the form used for the CM evaluations.



## 2.5 The period integral and the differential of the Mahler measure

For  $c \in \mathbb{C} \setminus K$  the denominator  $f - c$  is continuous and nonvanishing on  $\mathbb{T}^3$ , so the integral

$$\Omega(c) := \int_{\mathbb{T}^3} \frac{d\mu}{f(x, y, z) - c} \quad (21)$$

converges absolutely. Moreover  $\Omega$  is holomorphic on  $\mathbb{C} \setminus K$ : for any compact  $K_0 \subset \mathbb{C} \setminus K$  one has  $|f - c| \geq \text{dist}(K_0, K) > 0$  uniformly for  $c \in K_0$  and  $(x, y, z) \in \mathbb{T}^3$ , so differentiation under the integral sign is legitimate and gives  $\Omega'(c) = \int_{\mathbb{T}^3} d\mu / (f - c)^2$ . Being defined directly by the integral,  $\Omega$  is *single-valued* on  $\mathbb{C} \setminus K$ .

**Lemma 2.12** (Differential of the Mahler measure). *As real 1-forms on  $\mathbb{C} \setminus K$ ,*

$$d\mathfrak{m}(f - c) = \text{Re}[-\Omega(c) dc]. \quad (22)$$

*Consequently every branch of the holomorphic Mahler measure  $\tilde{\mathfrak{m}}$  of Lemma 2.2 satisfies  $\tilde{\mathfrak{m}}'(c) = -\Omega(c)$  on  $\mathbb{C} \setminus K$ : the derivative of  $\tilde{\mathfrak{m}}$  is single-valued, and any two branches differ by a locally constant pure imaginary constant. Moreover  $\Omega$  is odd:  $\Omega(-c) = -\Omega(c)$ .*

*Proof.* Write  $c = c_1 + ic_2$ . Since  $1/(f - c)$  is continuous on  $\mathbb{T}^3$  for  $c \notin K$ , we may differentiate under the integral sign:

$$\frac{\partial \mathfrak{m}(f - c)}{\partial c_1} = \int_{\mathbb{T}^3} \text{Re} \frac{-1}{f - c} d\mu = -\text{Re} \Omega(c), \quad \frac{\partial \mathfrak{m}(f - c)}{\partial c_2} = \int_{\mathbb{T}^3} \text{Re} \frac{-i}{f - c} d\mu = \text{Im} \Omega(c),$$

which is (22). If  $\tilde{\mathfrak{m}}$  is a local branch as in Lemma 2.2, then  $\text{Re} \tilde{\mathfrak{m}} = \mathfrak{m}(f - \cdot)$ , hence  $\text{Re}[\tilde{\mathfrak{m}}'(c) dc] = d\mathfrak{m}(f - c) = \text{Re}[-\Omega(c) dc]$ ; testing  $dc = 1$  and  $dc = i$  yields  $\tilde{\mathfrak{m}}'(c) = -\Omega(c)$ . For the oddness, substitute  $x \mapsto -x$  (Haar-invariant,  $f \mapsto -f$ ):  $\Omega(-c) = \int_{\mathbb{T}^3} d\mu / (f + c) = -\int_{\mathbb{T}^3} d\mu / (f - c) = -\Omega(c)$ .  $\square$

*Remark 2.13.* This is the precise sense in which the branch ambiguity of the holomorphic Mahler measure is invisible at the level of differentials: the multivaluedness of  $\tilde{\mathfrak{m}}$  lives in locally constant imaginary periods, while its derivative is the single-valued period integral (21) over the *fixed* integration cycle  $\mathbb{T}^3$ .

## 3 The continuation machine

This section develops the propagation argument used in the proof of Theorem A. It is stated for the family  $f + \sqrt{k}$  with Samart's parametrization  $k = s_2(\tau)$ ; Theorem B does not use it (see §5.1).

### 3.1 The propagation principle

**Lemma 3.1** (Propagation lemma). *Let  $U \subset \mathbb{C}$  be a connected open set, let  $Z \subset U$  be a relatively closed subset, and let  $H: U \rightarrow \mathbb{R}$  be continuous. Suppose that  $H$  is locally constant on  $U \setminus Z$ . Then  $H$  is constant on every connected component  $W$  of  $U \setminus Z$ . Moreover, if  $H \equiv C$  on  $W$ , then  $H \equiv C$  on  $\overline{W} \cap U$ .*

*Proof.* A locally constant function is constant on each connected component. For the second assertion, let  $p \in \overline{W} \cap U$  and choose  $p_n \in W$  with  $p_n \rightarrow p$ ; continuity gives  $H(p) = \lim_n H(p_n) = C$ .  $\square$

*Remark 3.2.* The conclusion cannot be upgraded to “ $H$  is constant on  $U$ ” in general: the devil’s staircase is continuous on  $[0, 1]$ , locally constant on the dense open complement of the Cantor set, yet non-constant. The membership of the target point in  $\overline{W}$  is therefore a genuine topological input; in our application it is supplied by the certified path of Proposition 4.9, and it does not come for free from the lemma.

### 3.2 The differential identity

Put

$$V := \{\tau \in \mathbb{H} : s_2(\tau) \notin [0, 64]\},$$

open since  $s_2$  is holomorphic. On  $V$ , the function

$$F(\tau) := n_2(s_2(\tau)) = 2m(f + \sqrt{s_2(\tau)})$$

is well defined (Lemma 2.1) and real-analytic: locally  $F(\tau) = 2m(f - c(\tau))$  for a holomorphic branch  $c(\tau) = \sqrt{s_2(\tau)}$ , which exists since  $s_2 \neq 0$  on  $V$ , and  $c(\tau) \notin [-8, 8]$  because  $c^2 = s_2(\tau) \notin [0, 64]$  (Remark 2.3); the value is independent of the branch since  $m(f + c) = m(f - c)$ . Also put

$$G(\tau) := EK(\tau) = 2 \operatorname{Re} E(\tau),$$

real-analytic on all of  $\mathbb{H}$ . By Lemma 2.1,  $F$  is continuous on all of  $\mathbb{H}$ .

**Lemma 3.3** (An open set in Samart’s region). *The disk  $D := \{\tau : |\tau - i| < \frac{1}{10}\}$  is contained in  $V \cap \{\operatorname{Im} \tau > \frac{1}{2}\}$ : indeed  $|s_2(\tau)| > 261$  on  $D$ .*

*Proof.* On  $D$  we have  $\operatorname{Im} \tau \geq 9/10$ , hence  $r := |q| \leq e^{-9\pi/5} < 0.0035$  and

$$|s_2(\tau)| = r^{-1} \prod_{n \geq 1} |1 + q^{2n-1}|^{24} \geq r^{-1} \left(1 - \sum_{n \geq 1} r^{2n-1}\right)^{24} = r^{-1} \left(1 - \frac{r}{1 - r^2}\right)^{24},$$

which is  $\geq e^{9\pi/5} (0.9965)^{24} > 261 > 64$ , so  $s_2(\tau) \notin [0, 64]$ .  $\square$

Let  $W$  be the connected component of  $V$  containing  $D$ .

**Proposition 3.4** (Differential identity). *There is a well-defined holomorphic function  $\Psi$  on  $V$  such that*

$$dF = 2 \operatorname{Re}[-\Psi(\tau) d\tau] + dG \quad \text{on } V, \tag{23}$$

*namely  $\Psi(\tau) = \Omega(c(\tau))c'(\tau) + E'(\tau)$  for any local holomorphic branch  $c(\tau) = \sqrt{s_2(\tau)}$ , the expression being independent of the branch. Moreover  $\Psi$  vanishes identically on  $W$ ; equivalently  $dF = dG$  on  $W$ .*

*Proof.* Branch independence: the other branch is  $-c$ , and by the oddness of  $\Omega$  (Lemma 2.12),  $\Omega(-c)(-c)' = (-\Omega(c))(-c') = \Omega(c)c'$ . Hence  $\Psi$  is a well-defined holomorphic function on  $V$ . The chain rule applied to (22) gives, locally,  $dF = 2 \operatorname{Re}[-\Omega(c(\tau))c'(\tau) d\tau]$ , and  $dG = 2 \operatorname{Re}[E'(\tau) d\tau]$ , which is (23).

On the nonempty open set  $D \subset W \cap \{\operatorname{Im} \tau > 1/2\}$  (Lemma 3.3), Theorem 2.9 gives  $F = G$ , hence  $dF = dG$ , i.e.  $\operatorname{Re}[\Psi(\tau) d\tau] = 0$  for all directions  $d\tau$ ; testing  $d\tau = 1$  and  $d\tau = i$  gives  $\Psi = 0$  on  $D$ . Both  $\Omega \circ c \cdot c'$  and  $E'$  are holomorphic on the connected open set  $W$ , so the identity theorem gives  $\Psi \equiv 0$  on  $W$ .  $\square$

### 3.3 The certified-path criterion

**Proposition 3.5** (Path criterion). *Let  $\tau_* \in \mathbb{H}$  and suppose there is a continuous path  $\gamma: [0, 1] \rightarrow \mathbb{H}$  with  $\gamma(0) \in W$ ,  $\gamma([0, 1)) \subset V$ , and  $\gamma(1) = \tau_*$ . Then  $\tau_* \in \overline{W}$  and*

$$n_2(s_2(\tau_*)) = \text{EK}(\tau_*).$$

*Proof.* Let  $H := F - G$ . By Lemma 2.1 and the real-analyticity of  $G$ ,  $H$  is continuous on all of  $\mathbb{H}$ ; by Proposition 3.4,  $dH = 0$  on  $W$ , so  $H$  is locally constant on  $W$ , hence constant on the connected set  $W$ ; on the nonempty open set  $D \subset W$ , Theorem 2.9 gives  $H = 0$ , so  $H \equiv 0$  on  $W$ . The set  $\gamma([0, 1))$  is a connected subset of  $V$  containing  $\gamma(0) \in W$ ; a connected subset of  $V$  meeting the component  $W$  lies entirely in  $W$ , so  $\gamma(t) \in W$  for  $t < 1$  and  $\tau_* = \lim_{t \rightarrow 1} \gamma(t) \in \overline{W}$ . Since  $H \equiv 0$  on  $W$  and  $H$  is continuous at  $\tau_*$ ,  $H(\tau_*) = \lim_{t \rightarrow 1} H(\gamma(t)) = 0$ .  $\square$

The topological hypothesis  $\gamma([0, 1)) \subset V$  is certified by interval arithmetic: the path is covered by finitely many parameter blocks, each carrying a strict interval enclosure of  $s_2$  certifiably disjoint from  $[0, 64]$  (Section 6). The endpoint  $\tau_*$ , where  $s_2(\tau_*)$  may lie on the cut  $[0, 64]$ , is absorbed by the continuity of Mahler measure — this is exactly where the genuine Mahler measure (as opposed to  $\text{Re } \tilde{m}$ ) enters.

## 4 Proof of Theorem A

Throughout this section

$$\tau_0 := \frac{3 + \sqrt{-7}}{8}, \quad 4\tau_0^2 - 3\tau_0 + 1 = 0, \quad \text{Im } \tau_0 = \frac{\sqrt{7}}{8}.$$

### 4.1 (P1) CM evaluation at $\tau_0$

Set

$$\beta := 4\tau_0 = \frac{3 + \sqrt{-7}}{2} = 1 + \omega.$$

Then  $N(\beta) = |\beta|^2 = 4$ , and  $\beta^2 = 3\beta - 4$  (from the quadratic equation for  $\tau_0$ ), whence  $\beta\omega = \beta(\beta - 1) = 2\beta - 4$ .

**Lemma 4.1.**  $\Lambda_4(\tau_0) = \mathcal{O}_K$  and  $\Lambda_1(\tau_0) = (\beta)/4$ , where  $(\beta) = \beta\mathcal{O}_K$  is the principal ideal of norm 4.

*Proof.*  $\Lambda_4(\tau_0) = \mathbb{Z} + \mathbb{Z}\beta = \mathbb{Z} + \mathbb{Z}(1 + \omega) = \mathbb{Z} + \mathbb{Z}\omega = \mathcal{O}_K$ . For the second,

$$(\beta) = \{\beta(m + n\omega) : m, n \in \mathbb{Z}\} = \{m\beta + n(2\beta - 4)\} = \{a\beta + 4b : a, b \in \mathbb{Z}\} = 4\mathbb{Z} + \mathbb{Z}\beta,$$

so  $(\beta)/4 = \mathbb{Z} + \mathbb{Z}(\beta/4) = \mathbb{Z} + \mathbb{Z}\tau_0 = \Lambda_1(\tau_0)$ .  $\square$

**Proposition 4.2.** With  $L_3 := L(g_7, 3)$ ,

$$T_4(\tau_0) = 4L_3 + 2\zeta_K(2), \quad T_1(\tau_0) = 8L_3 + 32\zeta_K(2). \quad (24)$$

*Proof.* For  $\Lambda_4(\tau_0) = \mathcal{O}_K$ , the alternative form (20) and Proposition 2.7 (equations (9) at  $s = 3$  and (10)) give immediately

$$T_4(\tau_0) = 2 \text{Re} \sum_{\alpha \in \mathcal{O}_K}' \frac{\bar{\alpha}^2}{|\alpha|^6} + \sum_{\alpha \in \mathcal{O}_K}' \frac{1}{|\alpha|^4} = 4L_3 + 2\zeta_K(2).$$

For  $\Lambda_1(\tau_0) = (\beta)/4$ , rescale  $\lambda = \alpha/4$ ,  $\alpha \in (\beta) \setminus \{0\}$ : then  $\bar{\lambda}^2/|\lambda|^6 = 256 \bar{\alpha}^2/|\alpha|^6$  and  $|\lambda|^{-4} = 256|\alpha|^{-4}$ . Writing  $\alpha = \beta\gamma$ ,  $\gamma \in \mathcal{O}_K \setminus \{0\}$ ,

$$\sum'_{\alpha \in (\beta)} |\alpha|^{-4} = |\beta|^{-4} \sum'_{\gamma \in \mathcal{O}_K} |\gamma|^{-4} = \frac{1}{16} \cdot 2\zeta_K(2),$$

$$\sum'_{\alpha \in (\beta)} \frac{\bar{\alpha}^2}{|\alpha|^6} = \frac{\bar{\beta}^2}{|\beta|^6} \sum'_{\gamma \in \mathcal{O}_K} \frac{\bar{\gamma}^2}{|\gamma|^6} = \frac{\bar{\beta}^2}{64} \cdot 2L_3,$$

since  $|\beta|^6 = (|\beta|^2)^3 = 64$ . Now  $\bar{\beta}^2 = ((3 - \sqrt{-7})/2)^2 = (1 - 3\sqrt{-7})/2$ , so  $\text{Re } \bar{\beta}^2 = 1/2$ . Therefore

$$T_1(\tau_0) = 256 \left[ 2 \cdot \frac{1}{2} \cdot \frac{2L_3}{64} + \frac{1}{16} \cdot 2\zeta_K(2) \right] = 8L_3 + 32\zeta_K(2). \quad \square$$

**Theorem 4.3** ((P1) at  $\tau_0$ ).

$$\text{EK}(\tau_0) = \frac{14\sqrt{7}}{\pi^3} L(g_7, 3) = 8M_7.$$

*Proof.* Combining (24),

$$-T_1(\tau_0) + 16T_4(\tau_0) = (-8 + 64)L_3 + (-32 + 32)\zeta_K(2) = 56L_3 :$$

the  $\zeta_K(2)$ -terms cancel *exactly* (the algebraic heart of the evaluation; the coefficients  $1/2 = \text{Re } \bar{\beta}^2$  and  $16 = |\beta|^4$  are responsible). By Lemma 2.10 with  $\text{Im } \tau_0 = \sqrt{7}/8$ ,

$$\text{EK}(\tau_0) = \frac{2 \cdot \sqrt{7}/8}{\pi^3} \cdot 56L_3 = \frac{14\sqrt{7}}{\pi^3} L_3 = 8 \cdot \frac{7\sqrt{7}}{4\pi^3} L_3 = 8M_7,$$

the last equality being (11).  $\square$

*Remark 4.4* (Numerical confirmation). The series  $T_1(\tau_0)$ ,  $T_4(\tau_0)$  computed directly by Poisson-summed rows (`verify_P1.py`, 60 dps) agree with (24) to 50 digits, and Theorem 4.3 holds numerically to  $5.6 \times 10^{-51}$ :

$$T_1(\tau_0) = 66.1354356158220540671143350434478014047047794968037983919065 \dots$$

$$T_4(\tau_0) = 6.5399375223477356279582944849702366138990879925962586125874 \dots$$

## 4.2 Exactness of $s_2(\tau_0) = 1$

**Lemma 4.5.**  $j(2\tau_0) = -3375$  *exactly*.

*Proof.*  $w := 2\tau_0 = (3 + \sqrt{-7})/4$  satisfies  $2w^2 - 3w + 2 = 0$ , a primitive quadratic form of discriminant  $-7$ . Since  $h(-7) = 1$ , the singular modulus  $j(w)$  is an algebraic integer lying in the Hilbert class field of  $K$ , which is  $K$  itself [12]. The reflected point  $-\bar{w} = (-3 + \sqrt{-7})/4 \in \mathbb{H}$  is a root of the primitive form  $2X^2 + 3X + 2$ , likewise of discriminant  $-7$ ; since  $h(-7) = 1$  there is a single class, so  $-\bar{w}$  is  $\text{SL}_2(\mathbb{Z})$ -equivalent to  $w$ . As  $j$  has rational  $q$ -expansion coefficients,  $\overline{j(w)} = j(-\bar{w}) = j(w)$ , so  $j(w)$  is real. A real algebraic integer in  $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{-7})/2]$  is an ordinary integer. Finally, interval evaluation (`cert0_s2_eq1.py`, using  $j = E_4^3/\Delta$  with rigorous tails) gives  $|j(2\tau_0) + 3375| < 10^{-66}$ , which pins the integer:  $j(2\tau_0) = -3375 = -15^3$ .  $\square$

**Lemma 4.6.**  $\lambda(2\tau_0) = (1 + 3\sqrt{-7})/2$  *exactly*.

*Proof.* The classical  $\lambda$ - $j$  relation states, identically,

$$j = 256 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(1 - \lambda)^2}.$$

By Lemma 4.5,  $\lambda(2\tau_0)$  is a root of

$$\Phi(X) := 256(X^2 - X + 1)^3 + 3375 X^2(1 - X)^2.$$

Exact factorization over  $\mathbb{Q}$  (verified by rational polynomial division in `cert0_s2_eq.1.py`):

$$\Phi(X) = (X^2 - X + 16) (256X^4 - 512X^3 + 303X^2 - 47X + 16), \quad (25)$$

and  $X^2 - X + 16$  divides  $\Phi$  with multiplicity one (the remainder of the second division is nonzero). The roots of  $X^2 - X + 16$  are  $(1 \pm \sqrt{-63})/2 = (1 \pm 3\sqrt{-7})/2$ . The four roots of the quartic factor are approximately  $31/32 \pm 0.24804i$  and  $1/32 \pm 0.24804i$ ; the minimum distance from  $\lambda_0 := (1 + 3\sqrt{-7})/2$  to any other root of  $\Phi$  is 3.75 (Appendix A). Since  $\Phi$  has rational coefficients and the minimal polynomial  $X^2 - X + 16$  of  $\lambda_0$  occurs as a simple factor, no other root equals  $\lambda_0$ . Interval evaluation gives  $|\lambda(2\tau_0) - \lambda_0| < 10^{-58} \ll 3.75$ , so  $\lambda(2\tau_0) = \lambda_0$ .  $\square$

**Corollary 4.7.**  $s_2(\tau_0) = 1$  *exactly*.

*Proof.*  $\lambda_0(1 - \lambda_0) = 16$  (since  $\lambda_0^2 - \lambda_0 + 16 = 0$ ), so by Theorem 2.5,  $s_2(\tau_0) = 16/(\lambda_0(1 - \lambda_0)) = 16/16 = 1$ .  $\square$

*Remark 4.8* (Weber’s viewpoint). The function  $s_2$  is a Weber class invariant in disguise: with Weber’s function (in his normalization)  $f_1(\tau) = \zeta_{48}^{-1} \eta((\tau + 1)/2)/\eta(\tau) = q^{-1/48} \prod_{n \geq 1} (1 + q^{n-1/2})$  one has  $s_2(\tau) = f_1(2\tau)^{24}$ , so Corollary 4.7 says that  $f_1$  assumes a 24th root of unity at  $2\tau_0$ . The value of Lemma 4.6 also has an arithmetic explanation. With  $\pi = (1 + \sqrt{-7})/2$  as in (2),

$$\lambda_0 = \pi^4, \quad 1 - \lambda_0 = \pi^4, \quad \lambda_0(1 - \lambda_0) = \pi^4 \bar{\pi}^4 = N(\pi)^4 = 2^4 = 16,$$

which is the source of the “16” in Theorem 2.5, and hence of  $s_2(\tau_0) = 1$ . Finally,  $j(2\tau_0) = -3375 = -15^3$  (Lemma 4.5) is the classical singular modulus of discriminant  $-7$ , tabulated already in Weber’s book [10] (with  $\gamma_2 = -15$ ); our interval-locked derivation above is independent of that table, whose known errata [11] make an independent check desirable.

### 4.3 The certified path

**Proposition 4.9** (Certified; see Section 6). *Let  $\gamma$  be the path from  $i$  to  $\tau_0$  consisting of*

- (i) *the vertical segment from  $i$  to  $9i/16$ ;*
- (ii) *the quarter circle  $\tau = \frac{i}{2} + \frac{1}{16}e^{i\theta}$ ,  $\theta$  from  $\pi/2$  to 0 (from  $9i/16$  to  $\frac{1}{16} + \frac{i}{2}$ );*
- (iii) *the horizontal segment from  $\frac{1}{16} + \frac{i}{2}$  to  $\frac{3}{8} + \frac{i}{2}$ ;*
- (iv) *the vertical segment from  $\frac{3}{8} + \frac{i}{2}$  to  $\tau_0$ .*

*Then  $\gamma \setminus \{\tau_0\} \subset V$ . Consequently  $\gamma \setminus \{\tau_0\} \subset W$  and  $\tau_0 \in \overline{W}$ .*

Indeed,  $\gamma \setminus \{\tau_0\}$  is a connected subset of  $V$  containing the starting point  $i \in D \subset W$  (Lemma 3.3); a connected subset of  $V$  meeting the component  $W$  lies entirely in  $W$ , and  $\gamma(t) \rightarrow \tau_0$  gives  $\tau_0 \in \overline{W}$ .

The inclusion  $\gamma \setminus \{\tau_0\} \subset V$  is certified by interval arithmetic in Section 6 (`cert2_path_k064.py`): each of the four pieces is covered by finitely many certified blocks (16, 18, 37, and  $15 + 13$  blocks respectively) whose  $s_2$ -enclosures are provably disjoint from  $[0, 64]$ , except for a final cap  $y \in (y_0, y_0 + \varepsilon_0]$ ,  $y_0 = \sqrt{7}/8$ ,  $\varepsilon_0 = 1/64000$ , which is closed analytically: interval evaluation on the box  $\{|\operatorname{Re} \tau - 3/8| \leq 0.03, |\operatorname{Im} \tau - y_0| \leq 0.03\} \subset \mathbb{H}$  gives the strict outward-rounded upper bound  $|s_2| \leq M_0$ ,  $M_0 = 38.148391199402133945$ ; by Cauchy's estimate, for  $|w - \tau_0| \leq 0.01$ ,

$$|s_2''(w)| \leq M_2 := 190742,$$

the strict interval value of  $2M_0/(0.02)^2$  being 190741.95599701066973. Since  $s_2(\tau_0) = 1$  *exactly* (Corollary 4.7),

$$s_2(\tau_0 + i\varepsilon) - 1 = i\varepsilon s_2'(\tau_0) + R, \quad |R| \leq M_2 \varepsilon^2/2,$$

and rigorous interval evaluation (the roundings of  $y_0$  and  $\varepsilon_0$  included inside the enclosures) yields

$$\operatorname{Re} s_2'(\tau_0) \in [-18.0197947489675, -15.0394516865142].$$

Therefore, for all  $\varepsilon \in (0, \varepsilon_0]$ ,

$$\operatorname{Im} s_2(\tau_0 + i\varepsilon) = \varepsilon \operatorname{Re} s_2'(\tau_0) + \operatorname{Im} R \leq \varepsilon(-15.0394\dots + M_2 \varepsilon_0/2) \leq -13.5492\dots \varepsilon < 0, \quad (26)$$

so  $s_2$  misses  $[0, 64]$  on the cap as well.  $\square$

#### 4.4 Conclusion

*Proof of Theorem A.* By Propositions 4.9 and 3.5 with  $\tau_* = \tau_0$ ,

$$n_2(s_2(\tau_0)) = \operatorname{EK}(\tau_0).$$

By Corollary 4.7,  $s_2(\tau_0) = 1$ , and  $n_2(1) = 2m(f+1) = 2m(f-1)$ . By Theorem 4.3,  $\operatorname{EK}(\tau_0) = 8M_7$ . Hence

$$2m(f+1) = \operatorname{EK}(\tau_0) = 8L'(g_7, 0). \quad \square$$

*Remark 4.10* (The  $\zeta_K(2)$ -cancellation). The exact cancellation of  $\zeta_K(2)$  in  $-T_1 + 16T_4$  is the arithmetic reason the  $k = 1$  evaluation exists in closed form. Its source is the pair of coincidences  $|\beta|^4 = 16$  and  $\operatorname{Re} \bar{\beta}^2 = 1/2$  for  $\beta = (3 + \sqrt{-7})/2$ . In the evaluation for Theorem B (§5.3) the analogous terms do *not* cancel, and their survival produces the Dirichlet term  $d_7$ .

## 5 Proof of Theorem B

Throughout this section

$$k_{\pm} := \frac{47 \pm 45\sqrt{-7}}{2}, \quad \tau' := \frac{1 + \sqrt{-7}}{8}, \quad \tau'' := -\overline{\tau'} = \frac{-1 + \sqrt{-7}}{8},$$

and the Fricke partners

$$\tau_w := -\frac{1}{4\tau'} = \frac{-1 + \sqrt{-7}}{4}, \quad \tau'_w := -\overline{\tau_w} = -\frac{1}{4\tau''} = \frac{1 + \sqrt{-7}}{4}.$$

Note  $|k_{\pm}|^2 = k_+ k_- = (47^2 + 45^2 \cdot 7)/4 = 16384/4 = 4096$ , i.e.  $|k_{\pm}| = 64$ , and  $\operatorname{Im} k_{\pm} = \pm 45\sqrt{7}/2 \neq 0$ , so  $k_{\pm} \notin [0, 64]$ . Also

$$\operatorname{Im} \tau' = \frac{\sqrt{7}}{8} < \frac{1}{2}, \quad \operatorname{Im} \tau_w = \operatorname{Im} \tau'_w = \frac{\sqrt{7}}{4} \geq \frac{1}{2}. \quad (27)$$

## 5.1 The Fricke lemma and the two preimages

**Lemma 5.1** (Fricke invariance of  $s_2$ ).  $s_2(-1/(4\tau)) = s_2(\tau)$  for all  $\tau \in \mathbb{H}$ .

*Proof.* By Theorem 2.5,  $s_2(\tau) = 16/(\lambda(2\tau)(1 - \lambda(2\tau)))$ . The classical transformation  $\lambda(-1/u) = 1 - \lambda(u)$  with  $u = 2\tau$  gives  $\lambda(2 \cdot (-1/(4\tau))) = \lambda(-1/(2\tau)) = 1 - \lambda(2\tau)$ ; since  $x(1 - x)$  is invariant under  $x \mapsto 1 - x$ , the claim follows. (Numerically confirmed at sample points to  $10^{-57}$  in `verify_P1_n2pair.py`.)  $\square$

Since  $s_2$  has degree 2 on  $X_0(4)$  (Remark 2.6),  $\tau'$  and  $\tau_w = -1/(4\tau')$  are the two preimages of  $k_+$  relevant here.

## 5.2 Exactness of $s_2(\tau') = (47 + 45\sqrt{-7})/2$

**Lemma 5.2.**  $j(2\tau') = -3375$  exactly — the same value as  $j(2\tau_0)$  (Lemma 4.5).

*Proof.*  $w := 2\tau' = (1 + \sqrt{-7})/4 = \pi/2$  satisfies  $2w^2 - w + 1 = 0$ , a primitive quadratic form of discriminant  $-7$ . Since  $h(-7) = 1$ , exactly as in Lemma 4.5,  $j(w)$  is a real algebraic integer in  $\mathcal{O}_K$ , hence an ordinary integer. Interval evaluation (`cert0_s2_n2pair.py`, 100 dps, rigorous tails) gives  $|j(2\tau') + 3375| \leq 1.8 \times 10^{-91}$ , so  $j(2\tau') = -3375$ .  $\square$

*Remark 5.3* (A structural coincidence). Both  $2\tau_0 = (3 + \sqrt{-7})/4$  and  $2\tau' = (1 + \sqrt{-7})/4$  are roots of primitive positive definite quadratic forms of discriminant  $-7$  ( $2X^2 - 3X + 2$  and  $2X^2 - X + 1$  respectively). Since  $h(-7) = 1$  there is a single singular modulus of discriminant  $-7$ , namely  $-3375 = -15^3$  [10, 12]: the same sextic  $\Phi(X) = 256(X^2 - X + 1)^3 + 3375X^2(1 - X)^2$  therefore governs the  $\lambda$ -values in both theorems.

**Lemma 5.4.**  $\lambda(2\tau') = (31 + 3\sqrt{-7})/32$  exactly.

*Proof.* By Lemma 5.2,  $\lambda(2\tau')$  is a root of the same sextic  $\Phi$  of (25). Over  $\mathbb{Q}(\sqrt{-7})$  the sextic splits completely into three quadratic factors (exact factorization, verified by rational polynomial division in `cert0_s2_n2pair.py`):

$$\Phi(X) = (X^2 - X + 16)(16X^2 - X + 1)(16X^2 - 31X + 16), \quad (28)$$

with the six roots

$$\frac{1 \pm 3\sqrt{-7}}{2}, \quad \frac{1 \pm 3\sqrt{-7}}{32}, \quad \frac{31 \pm 3\sqrt{-7}}{32}$$

(Appendix A). The candidate  $\lambda_1 := (31 + 3\sqrt{-7})/32$  is a root of the third factor  $16X^2 - 31X + 16$ ; its minimal distance to the other five roots of  $\Phi$  is  $3\sqrt{7}/16 = 0.4960\dots$  (attained at its conjugate). Interval evaluation gives  $|\lambda(2\tau') - \lambda_1| \leq 1.2 \times 10^{-62} \ll 0.496$ , so  $\lambda(2\tau') = \lambda_1$ .  $\square$

**Corollary 5.5.**  $s_2(\tau') = k_+$  and  $s_2(\tau'') = k_-$  exactly; consequently  $s_2(\tau_w) = k_+$  and  $s_2(\tau'_w) = k_-$  exactly.

*Proof.* From  $16\lambda_1^2 - 31\lambda_1 + 16 = 0$  we get  $\lambda_1(1 - \lambda_1) = (16 - 15\lambda_1)/16$ , and substituting  $\lambda_1 = (31 + 3\sqrt{-7})/32$ ,

$$\lambda_1(1 - \lambda_1) = \frac{47 - 45\sqrt{-7}}{512}.$$

By Theorem 2.5,

$$s_2(\tau') = \frac{16}{\lambda_1(1 - \lambda_1)} = \frac{8192}{47 - 45\sqrt{-7}} = \frac{47 + 45\sqrt{-7}}{2} = k_+,$$

since  $(47 - 45\sqrt{-7})(47 + 45\sqrt{-7}) = 47^2 + 7 \cdot 45^2 = 16384 = 2 \cdot 8192$ . Since  $s_2$  has integral (hence real)  $q$ -coefficients,  $s_2(-\bar{\tau}) = \overline{s_2(\tau)}$ , and  $\tau'' = -\bar{\tau}'$  gives  $s_2(\tau'') = k_-$ . The Fricke lemma (Lemma 5.1) then gives  $s_2(\tau_w) = s_2(\tau') = k_+$  and  $s_2(\tau'_w) = s_2(\tau'') = k_-$ .  $\square$

### 5.3 (P1) CM evaluation at $\tau_w$

Recall  $\pi = (1 + \sqrt{-7})/2$  and the relations (2). Since  $4\tau_w = -1 + \sqrt{-7} = -2\bar{\pi}$ ,

$$\Lambda_4(\tau_w) = \mathbb{Z} + \mathbb{Z} 4\tau_w = \mathbb{Z} + \mathbb{Z} 2\bar{\pi} = \mathbb{Z} + \mathbb{Z}\sqrt{-7} =: \mathcal{O}_2, \quad (29)$$

the order of conductor 2 in  $\mathcal{O}_K$  (discriminant  $-28$ , class number 1 — only its  $\mathbb{Z}$ -module structure is used below), and

$$\Lambda_1(\tau_w) = \mathbb{Z} + \mathbb{Z}\tau_w = \frac{4\mathbb{Z} + \mathbb{Z} 4\tau_w}{4} = \frac{2\mathbb{Z} + \mathbb{Z}\bar{\pi}}{2} = \frac{(\bar{\pi})}{2}, \quad (30)$$

because  $(\bar{\pi}) = \bar{\pi}\mathcal{O}_K = \bar{\pi}\mathbb{Z} + \bar{\pi}^2\mathbb{Z} = \bar{\pi}\mathbb{Z} + (\bar{\pi} - 2)\mathbb{Z} = 2\mathbb{Z} + \bar{\pi}\mathbb{Z}$  (using  $\bar{\pi}^2 = \bar{\pi} - 2$ ). So  $\Lambda_1(\tau_w)$  is homothetic to the principal ideal  $(\bar{\pi})$  of norm 2. At the conjugate point  $\tau'_w = \pi/2$  one gets  $\Lambda_4(\tau'_w) = \mathcal{O}_2$  (the same) and  $\Lambda_1(\tau'_w) = (\pi)/2$  — the other prime above 2.

**Imprimitive sums over  $\mathcal{O}_2$ .** Define

$$B(\mathcal{O}_2) := \sum'_{\gamma \in \mathcal{O}_2} |\gamma|^{-4}, \quad G(\mathcal{O}_2) := \sum'_{\gamma \in \mathcal{O}_2} \frac{\bar{\gamma}^2}{|\gamma|^6}.$$

Note  $G(\mathcal{O}_2)$  is real:  $\mathcal{O}_2 = \mathbb{Z} + \mathbb{Z}\sqrt{-7}$  is stable under conjugation, so the sum equals its own conjugate.

**Lemma 5.6** (Odd norm  $\Leftrightarrow 1 \pmod{2}$ ). *For  $\gamma = a + b\pi \in \mathcal{O}_K$ ,  $N(\gamma) = a^2 + ab + 2b^2 \equiv a(a + b) \pmod{2}$ . Hence  $N(\gamma)$  is odd if and only if  $a$  is odd and  $b$  is even, i.e.  $\gamma \equiv 1 \pmod{2\mathcal{O}_K}$ .*

*Proof.*  $a^2 + ab + 2b^2 \equiv a^2 + ab = a(a + b) \pmod{2}$ , and  $a(a + b)$  is odd iff  $a$  is odd and  $a + b$  is odd, i.e.  $a$  odd,  $b$  even.  $\square$

**Corollary 5.7** (Decomposition).  $\mathcal{O}_2 \setminus \{0\} = (2\mathcal{O}_K \setminus \{0\}) \sqcup \{\gamma \in \mathcal{O}_K : N(\gamma) \text{ odd}\}.$

*Proof.*  $\mathcal{O}_2 = \mathbb{Z} + \mathbb{Z}\sqrt{-7} = \mathbb{Z} + \mathbb{Z}(1 + \sqrt{-7}) = \mathbb{Z} + 2\mathbb{Z}\pi = \mathbb{Z} + 2\mathcal{O}_K$ , so  $\gamma \in \mathcal{O}_2$  is either in  $2\mathcal{O}_K$  or in  $1 + 2\mathcal{O}_K$ , and Lemma 5.6 applies.  $\square$

**Proposition 5.8** (Evaluation of the imprimitive sums).

$$B(\mathcal{O}_2) = \frac{5}{4} \zeta_K(2), \quad G(\mathcal{O}_2) = 3 L(g_7, 3).$$

*Proof.* By Lemma 5.6, every ideal  $\mathfrak{a} \triangleleft \mathcal{O}_K$  of odd norm has both its generators  $(\pm\gamma, \text{ and } -1 \equiv 1 \pmod{2})$  in the ray class 1 mod 2, so for  $\text{Re } s$  large

$$\sum'_{N(\gamma) \text{ odd}} N(\gamma)^{-s} = 2 \sum'_{N(\mathfrak{a}) \text{ odd}} N(\mathfrak{a})^{-s} = 2 \zeta_K(s) (1 - N(\pi)^{-s}) (1 - N(\bar{\pi})^{-s}) = 2 \zeta_K(s) (1 - 2^{-s})^2. \quad (31)$$

At  $s = 2$ :  $\sum'_{N(\gamma) \text{ odd}} |\gamma|^{-4} = 2 \zeta_K(2) \cdot (3/4)^2 = (9/8) \zeta_K(2)$ , hence by Corollary 5.7 and (10),

$$B(\mathcal{O}_2) = \sum_{\gamma \in 2\mathcal{O}_K \setminus \{0\}} |\gamma|^{-4} + \sum'_{N(\gamma) \text{ odd}} |\gamma|^{-4} = 2^{-4} \cdot 2 \zeta_K(2) + \frac{9}{8} \zeta_K(2) = \frac{5}{4} \zeta_K(2).$$



For the  $\psi$ -twisted sum,  $\psi((\alpha)) = \alpha^2$  contributes equally for both generators  $((-\alpha)^2 = \alpha^2)$ , so by (9),

$$\sum_{N(\gamma) \text{ odd}}' \frac{\gamma^2}{N(\gamma)^3} = 2L(g_7, 3) \left(1 - \frac{\pi^2}{8}\right) \left(1 - \frac{\bar{\pi}^2}{8}\right) = 2L(g_7, 3) \cdot \frac{23}{16} = \frac{23}{8} L(g_7, 3), \quad (32)$$

the Euler factor being *exact* by (2):

$$\left(1 - \frac{\pi^2}{8}\right) \left(1 - \frac{\bar{\pi}^2}{8}\right) = 1 - \frac{\pi^2 + \bar{\pi}^2}{8} + \frac{(\pi\bar{\pi})^2}{64} = 1 + \frac{3}{8} + \frac{1}{16} = \frac{23}{16}.$$

Conjugating (note  $L(g_7, 3) \in \mathbb{R}$  since  $a_n \in \mathbb{Z}$ ) gives the same value with  $\bar{\gamma}^2$ , and

$$G(\mathcal{O}_2) = \sum_{\gamma \in 2\mathcal{O}_K \setminus \{0\}} \frac{\bar{\gamma}^2}{|\gamma|^6} + \sum_{N(\gamma) \text{ odd}}' \frac{\bar{\gamma}^2}{|\gamma|^6} = \frac{4}{64} \cdot 2L(g_7, 3) + \frac{23}{8} L(g_7, 3) = 3L(g_7, 3). \quad \square$$

*Remark 5.9* (Independent cross-checks). (i) Direct Poisson/Bessel Epstein evaluation of  $B(\mathcal{O}_2)$ ,  $G(\mathcal{O}_2)$  (the rectangular lattice form  $u^2 + 7v^2$ ) agrees with Proposition 5.8 to  $10^{-61}$ . (ii) The Glasser–Zucker closed form for the discriminant- $(-28)$  Epstein zeta [13],  $Z(\mathcal{O}_2, s) = 2(1 - 2^{1-s} + 2^{1-2s}) \zeta(s) L(\chi_{-7}, s)$ , at  $s = 2$  gives  $2(1 - \frac{1}{2} + \frac{1}{8}) \zeta_K(2) = \frac{5}{4} \zeta_K(2)$ , agreeing with  $B(\mathcal{O}_2)$  — an independent literature formula. (iii) The complementary coset sums over  $\pi + \mathcal{O}_2$  (since  $\mathcal{O}_K = \mathcal{O}_2 \sqcup (\pi + \mathcal{O}_2)$ ) evaluate to  $B(\mathcal{O}_K) - B(\mathcal{O}_2) = (3/4) \zeta_K(2)$  and  $2L(g_7, 3) - 3L(g_7, 3) = -L(g_7, 3)$ , confirmed directly to  $10^{-61}$  (`verify_P1_n2pair.py`, checks [O1]–[O3]).

**Proposition 5.10** (The  $T$ -sums at  $\tau_w$ ). *With  $L_3 := L(g_7, 3)$ ,*

$$T_4(\tau_w) = 6L_3 + \frac{5}{4} \zeta_K(2), \quad T_1(\tau_w) = -12L_3 + 8\zeta_K(2). \quad (33)$$

*The same values hold at  $\tau'_w$ .*

*Proof.* By (20) with  $\Lambda_4(\tau_w) = \mathcal{O}_2$  and Proposition 5.8,

$$T_4(\tau_w) = 2 \operatorname{Re} G(\mathcal{O}_2) + B(\mathcal{O}_2) = 6L_3 + \frac{5}{4} \zeta_K(2).$$

For  $\Lambda_1(\tau_w) = (\bar{\pi})/2$ , write  $\lambda = \alpha/2$ ,  $\alpha \in (\bar{\pi}) \setminus \{0\}$ ,  $\alpha = \bar{\pi}\gamma$ ,  $\gamma \in \mathcal{O}_K \setminus \{0\}$ . Then  $|\lambda|^{-4} = 16|\alpha|^{-4}$  and  $\bar{\lambda}^2/|\lambda|^6 = 16\bar{\alpha}^2/|\alpha|^6$ , with

$$\sum_{\alpha \in (\bar{\pi})}' |\alpha|^{-4} = |\bar{\pi}|^{-4} \sum_{\gamma \in \mathcal{O}_K}' |\gamma|^{-4} = N(\bar{\pi})^{-2} \cdot 2\zeta_K(2) = \frac{\zeta_K(2)}{2},$$

$$\sum_{\alpha \in (\bar{\pi})}' \frac{\bar{\alpha}^2}{|\alpha|^6} = \frac{\pi^2}{|\bar{\pi}|^6} \sum_{\gamma \in \mathcal{O}_K}' \frac{\bar{\gamma}^2}{|\gamma|^6} = \frac{\pi^2}{8} \cdot 2L_3 = \frac{\pi^2 L_3}{4},$$

so, using  $\operatorname{Re} \pi^2 = -3/2$  from (2),

$$T_1(\tau_w) = 16 \left[ 2 \operatorname{Re} \frac{\pi^2 L_3}{4} + \frac{\zeta_K(2)}{2} \right] = 16 \left[ -\frac{3}{4} L_3 + \frac{\zeta_K(2)}{2} \right] = -12L_3 + 8\zeta_K(2).$$

At  $\tau'_w$  the same computation with  $(\pi)$  in place of  $(\bar{\pi})$  gives  $\operatorname{Re} \bar{\pi}^2 = -3/2$  — the same value; hence  $T_d(\tau'_w) = T_d(\tau_w)$  exactly.  $\square$

**Theorem 5.11** ((P1) at  $\tau_w$ ).

$$\text{EK}(\tau_w) = \frac{4}{7} (54M_7 + d_7).$$

*Proof.* Combining (33),

$$-T_1(\tau_w) + 16T_4(\tau_w) = (12 + 96)L_3 + (-8 + 20)\zeta_K(2) = 108L_3 + 12\zeta_K(2). \quad (34)$$

Note the coefficient of  $\zeta_K(2)$  is  $12 \neq 0$ : unlike the  $k = 1$  case, the  $\zeta_K(2)$ -terms do *not* cancel — this is precisely where the  $d_7$ -term of the final formula comes from. With  $\text{Im } \tau_w = \sqrt{7}/4$ , Lemma 2.10 gives

$$\begin{aligned} \text{EK}(\tau_w) &= \frac{\sqrt{7}}{2\pi^3} (108L(g_7, 3) + 12\zeta_K(2)) \\ &= \frac{\sqrt{7}}{2\pi^3} \cdot 108 \cdot \frac{4\pi^3}{7\sqrt{7}} M_7 + \frac{\sqrt{7}}{2\pi^3} \cdot 12 \cdot \frac{2\pi^3}{21\sqrt{7}} d_7 \quad [\text{by (11) and (13)}] \\ &= \frac{216}{7} M_7 + \frac{4}{7} d_7 = \frac{4}{7} (54M_7 + d_7). \end{aligned} \quad \square$$

## 5.4 Conclusion

*Proof of Theorem B.* By (27),  $\text{Im } \tau_w = \sqrt{7}/4 \geq 1/2$ , so Samart's identity (16) applies *directly* at  $\tau_w$  — no continuation argument is needed. By Corollary 5.5 and Theorem 5.11,

$$n_2(k_+) = n_2(s_2(\tau_w)) = \text{EK}(\tau_w) = \frac{4}{7} (54M_7 + d_7).$$

Likewise  $n_2(k_-) = n_2(s_2(\tau'_w)) = \text{EK}(\tau'_w) = \text{EK}(\tau_w)$  (Proposition 5.10), or simply  $n_2(k_-) = n_2(\overline{k_+}) = n_2(k_+)$  since  $f$  has real coefficients.  $\square$

*Remark 5.12* (The wrong-sheet companion at  $\tau'$ ). Samart's table [3] attaches the conjecture for  $k_{\pm}$  to the CM points  $\tau'$ ,  $\tau''$ , whose imaginary part  $\sqrt{7}/8$  is *below*  $1/2$ ; Theorem 2.9 does not apply there, and indeed the identity *fails* at  $\tau'$ . Numerically (60 dps, `diag_ek_branch.py` and check [WS] of `verify_P1_n2pair.py`), direct Poisson-row evaluation of the lattice sums at  $\tau'$  (where  $\Lambda_4(\tau') = \mathbb{Z} + \mathbb{Z}\pi = \mathcal{O}_K$  but  $\Lambda_1(\tau') = \pi\mathcal{O}_2/4$  is an  $\mathcal{O}_2$ -module *not* homothetic to an  $\mathcal{O}_K$ -ideal) gives

$$T_4(\tau') = 4L(g_7, 3) + 2\zeta_K(2), \quad T_1(\tau') = -288L(g_7, 3) + 80\zeta_K(2),$$

hence

$$\text{EK}(\tau') = \frac{\sqrt{7}}{4\pi^3} (352L(g_7, 3) - 48\zeta_K(2)) = \frac{8}{7} (44M_7 - d_7) = 3.22268\dots \neq 4.13826\dots = n_2(s_2(\tau')),$$

a difference of  $0.91558\dots$ . The function  $\text{EK}$  is single-valued real-analytic on all of  $\mathbb{H}$ , while  $\tau \mapsto n_2(s_2(\tau))$  is real-analytic only on  $V = \{\tau : s_2(\tau) \notin [0, 64]\}$ ; numerical scans (`diag_ek_branch.py`) show that paths from the imaginary axis to  $\tau'$  cross the locus  $\{\text{Im } s_2 = 0, 0 < \text{Re } s_2 < 64\}$  (at  $\text{Re } s_2 \approx 46.98$ ) — the pullback of the branch cut  $[0, 64]$  of the Rogers branch (3) — after which  $\text{EK}$  continues on a companion sheet whose value is exactly  $(8/7)(44M_7 - d_7)$ . This is the concrete sense in which  $m = \text{Re } \tilde{m}$  fails to be automatic at interior parameters, and it is precisely the obstruction that the certified-path continuation of Theorem A overcomes: the  $s_2$ -image of the path of Proposition 4.9 is certified to avoid  $[0, 64]$  entirely — through  $\text{Re } s_2 > 64$  on the axis piece, and through  $\text{Im } s_2 < 0$  on the arc, horizontal, vertical, and cap pieces.

The same mechanism is even more striking in Samart’s  $n_4$ -family  $n_4(s) = 4m(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz)$ : at  $s = 81$  the conjectured value  $40M_7$  [3, Table 6] coincides with the exact series evaluation at the attached CM point on a wrong sheet, while direct torus integration gives  $n_4(81) = 4.1655349907533676508(5)$  — the discrepancy  $+0.05867\dots$  is  $2 \times 10^5$  times the integration error floor, so the literal identity is numerically refuted; see the companion note [2] for the exact series-side evaluation and the obstruction analysis.

## 6 Machine certification and reproducibility

The non-elementary computations of this paper are certified by interval arithmetic. The object entering the proofs is not a program but a *certificate*: a finite list of parameter blocks, each carrying a strict complex-interval enclosure of  $s_2$  on the whole block, whose validity is checked by deterministic, finite, and independently re-runnable interval evaluations. The trusted base of the certification consists of exactly three items: (i) the tail bound of Lemma 6.2, a purely mathematical statement; (ii) the documented outward-rounding semantics of the interval arithmetic employed (`mpmath.iv` rounds all interval endpoints outward, with guard digits for the complex transcendental functions [15, 16]); and (iii) the certificate data itself, emitted by the scripts in machine-readable form. Two implementation safeguards support (ii) and (iii), without belonging to the trusted base themselves. First, the certificate path of `cert2_path_k064.py` (and of its companion `cert2_path.py`) contains no machine-`float` arithmetic: every bound is produced by outward-rounded interval operations, and all constants entering the certificates ( $\sqrt{7}/8$ ,  $1/64000$ ,  $\pi/2$ , ...) are themselves enclosed by intervals. Second, each script runs a self-test in which independent 80-dps high-precision point values are verified to lie inside the certified block enclosures (20/20 sample points contained; point-enclosures contained in block-enclosures 5/5) — an empirical corroboration of (ii), not a premise of the proof. The operative comparison throughout is *certified enclosure width versus certified distance to the forbidden set*  $[0, 64]$ , never a count of stable decimal digits. All scripts are plain Python 3.12 + `mpmath` [15] (standard library `fractions` for the exact algebra), run independently of one another.

### 6.1 Rigorous enclosure of $s_2$

For  $\tau$  in a (rectangular complex) interval, enclose  $q = e^{2\pi i\tau}$  by intervals, set  $r := \overline{|q|}$  (a certified upper bound), truncate the product (5) at  $N = 120$ , and attach a rigorous tail factor. The logical basis for trusting such an evaluation is the inclusion property of interval arithmetic [17], which we record explicitly:

**Lemma 6.1** (Inclusion property). *Let  $E$  be any expression built from interval-enclosed constants, the variable, interval arithmetic operations, and interval elementary functions, all with outward-rounded semantics. Then for every interval box  $I$  in its domain, the interval evaluation  $E(I)$  contains the true value  $E(\tau)$  for every  $\tau \in I$ .*

*Proof.* Induction on the structure of  $E$ . Constants and the variable are enclosed by definition (every constant such as  $\sqrt{7}/8$ ,  $\pi$ ,  $1/16$  enters as an enclosing interval, and  $\tau \in I$ ). If  $E = g(E_1, \dots, E_r)$  and  $E_i(\tau) \in E_i(I)$  by induction, then the outward-rounded interval operation  $g$  applied to the boxes  $E_i(I)$  returns a box containing  $g(z_1, \dots, z_r)$  for all  $z_i \in E_i(I)$ , in particular containing  $E(\tau)$ . (This is the fundamental theorem of interval analysis; cf. [17].)  $\square$

Lemma 6.1 reduces the correctness of any single interval *evaluation* to the outward-rounding semantics of the individual operations, i.e. to item (ii) of the trusted base. The remaining ingredient

is the truncation error, controlled mathematically:

**Lemma 6.2** (Tail bound). *If  $r < 1$  then*

$$\prod_{n>N} (1 + q^{2n-1})^{24} \in \exp([-E, E]), \quad E := \frac{24 r^{2N+1}}{(1-r)(1-r^2)}.$$

*Proof.* For  $|z| \leq r < 1$ ,  $|\log(1+z)| \leq -\log(1-|z|) \leq |z|/(1-r)$ . Summing,

$$\left| \sum_{n>N} 24 \log(1 + q^{2n-1}) \right| \leq \frac{24}{1-r} \sum_{n>N} r^{2n-1} = \frac{24 r^{2N+1}}{(1-r)(1-r^2)}. \quad \square$$

On the path of Proposition 4.9 the imaginary part is at least  $\sqrt{7}/8 + \varepsilon_0$  on every certified piece (attained at the bottom of the sub-cap), so  $r \leq e^{-2\pi(\sqrt{7}/8 + \varepsilon_0)} < 0.1252$  and with  $N = 120$  the tail width satisfies  $E < 10^{-215}$ . The scripts in fact attach the slightly stronger complex-box tail factor  $\exp(\{u + iv : |u|, |v| \leq E\})$ , bounding the argument of the tail as well as its modulus, with all quantities entering  $E$  outward-rounded intervals; this is used both for the path certification and for the interval locks of Lemmas 4.6 and 5.4.

A complex interval  $Z$  is *certified to miss*  $[0, 64]$  iff  $0 \notin \text{Im } Z$  or  $\text{Re } Z \cap [0, 64] = \emptyset$ . If the enclosure of a parameter block is certified to miss  $[0, 64]$ , then the entire block lies in  $V$ : this is a proof, not an estimate.

## 6.2 The certificate and its correctness

A *path certificate* is a finite list of parameter blocks  $I_1, \dots, I_n$  covering the path down to the endpoint cap, together with complex intervals  $Z_1, \dots, Z_n$  such that  $s_2(I_\nu) \subset Z_\nu$  and  $Z_\nu \cap [0, 64] = \emptyset$  for every  $\nu$ . The certificate for the path of Proposition 4.9 consists of 99 blocks ( $y_0 = \sqrt{7}/8$ ,  $\varepsilon_0 = 1/64000$ ):

| piece       | parameter range                                                       | certified in       |
|-------------|-----------------------------------------------------------------------|--------------------|
| axis        | $x = 0, y \in [9/16, 1]$                                              | 16 blocks          |
| quarter arc | $\tau = \frac{i}{2} + \frac{1}{16}e^{i\theta}, \theta \in [0, \pi/2]$ | 18 blocks          |
| horizontal  | $x \in [1/16, 3/8], y = 1/2$                                          | 37 blocks          |
| vertical    | $x = 3/8, y \in [y_0 + 10^{-3}, 1/2]$                                 | 15 blocks          |
| sub-cap     | $x = 3/8, y \in [y_0 + \varepsilon_0, y_0 + 10^{-3}]$                 | 13 blocks          |
| cap         | $x = 3/8, y \in (y_0, y_0 + \varepsilon_0]$                           | analytic, see (26) |

The complete list of blocks and enclosures is emitted by `cert2_path_k064.py` in machine-readable form; only the most dangerous block of each piece is reproduced in the margin table below.

**Proposition 6.3** (Correctness of the certificate). *Assume (i) the tail bound of Lemma 6.2 and (ii) that every interval operation used returns an enclosure of the true value (the outward-rounding semantics of [16]). Then each listed block satisfies*

$$s_2(I_\nu) \subset Z_\nu, \quad Z_\nu \cap [0, 64] = \emptyset.$$

*Consequently the covered part of the path lies in  $V$ .*

*Proof.* By Lemma 6.1 (applicable by (ii)), the interval evaluation of  $q$ , of each truncated factor  $(1 + q^{2n-1})^{24}$ , and of their product returns an enclosure of the corresponding true value on  $I_\nu$ ; by (i), multiplication by the complex-box tail factor enlarges this to an enclosure  $Z_\nu$  of the full

infinite product  $s_2$  on  $I_\nu$ . The disjointness  $Z_\nu \cap [0, 64] = \emptyset$  is decided from the endpoints of  $Z_\nu$  by the certified-miss predicate of the preceding subsection ( $0 \notin \text{Im } Z_\nu$  or  $\text{Re } Z_\nu \cap [0, 64] = \emptyset$ ), again using only (ii). The verification of the certificate is therefore a deterministic finite computation, independent of how the blocks were found: the PASSED output of `cert2_path_k064.py` is one execution of it, and any independent outward-rounded interval implementation re-executing the same block list obtains the same verdicts.  $\square$

The endpoint cap is *not* part of the block certificate; it is the analytic supplement of Section 4, closed by the Taylor argument of (26), which uses the *exact* value  $s_2(\tau_0) = 1$ , the Cauchy bound  $M_2 = 190742$  derived from the certified bound  $M_0 = 38.148391199402133945$  on a 0.03-box, and the certified enclosure  $\text{Re } s'_2(\tau_0) \in [-18.0197947489675, -15.0394516865142]$ .

For each piece, the most dangerous accepted block (the one whose enclosure comes closest to  $[0, 64]$ ) is:

| piece       | worst enclosure $Z$                                                                                  | width                | dist( $Z, [0, 64]$ )   |
|-------------|------------------------------------------------------------------------------------------------------|----------------------|------------------------|
| axis        | $\text{Re} \in [64.65, 75.59], \text{Im} \ni 0$                                                      | 10.94                | 0.648                  |
| quarter arc | $\text{Re} \in [55.02, 66.03], \text{Im} \subset [-5.098, -0.069]$                                   | 11.01                | 0.0690                 |
| horizontal  | $\text{Re} \in [58.62, 60.09], \text{Im} \subset [-1.094, -0.0461]$                                  | 1.471                | 0.0461                 |
| vertical    | $\text{Re} \in [1.0016, 1.1115], \text{Im} \subset [-0.0983, -0.0051]$                               | 0.11                 | $5.08 \times 10^{-3}$  |
| sub-cap     | $\text{Re} \in [1.00005, 1.00067], \text{Im} \subset [-5.836 \times 10^{-4}, -6.009 \times 10^{-5}]$ | $6.2 \times 10^{-4}$ | $6.009 \times 10^{-5}$ |

On the axis piece the separation is one-sided through  $\text{Re } Z > 64$  (the values are real there); on the arc, horizontal, vertical, and sub-cap pieces it is through  $\text{Im } Z < 0$  (the  $s_2$ -image passes below the cut). The global minimum margin over all 99 blocks is  $6.009249 \times 10^{-5} > 0$  (attained on the sub-cap), while the widest enclosure (axis piece, width 1022) stays at distance 77.7 from  $[0, 64]$ .

*Remark 6.4* (Generation and termination of the search). The block list was found by adaptive bisection: a block is accepted when its enclosure is certified to miss  $[0, 64]$ , otherwise it is bisected. This search is not part of the trusted base (Proposition 6.3), but it is worth recording why it had to terminate. By the standard contraction principle for interval extensions [17], the natural interval extension of an expression built from operations that are Lipschitz and non-singular on a box has width  $O(\text{diam})$ . The expression used here —  $q = e^{2\pi i \tau}$ , the factors  $(1 + q^{2n-1})^{24}$  with  $|q| < 0.1252 < 1$  and  $|1 + q^{2n-1}| \geq 1 - |q| > 0$  uniformly, and the tail factor of Lemma 6.2 (width  $< 10^{-215}$  uniformly) — satisfies these hypotheses on every piece, so enclosure widths shrink linearly with the block diameter. On each compact piece the true image has positive distance to the forbidden set: through  $\text{Im } s_2 < 0$  on the arc, horizontal, vertical, and sub-cap pieces, and through the one-sided separation  $\text{Re } s_2 > 64$  on the axis piece, where the values are real. Once the block diameter is small enough that the enclosure width is below that margin, the block is accepted; bisection therefore accepts after finitely many steps.

*Remark 6.5* (Relation to the companion script). The companion script `cert2_path.py` certifies the shorter path  $i/2 \rightarrow 3/8 + i/2 \rightarrow \tau_0$  with the avoided set  $[-8, 8]$  in place of  $[0, 64]$  (5, 15, and 13 blocks respectively, plus the same cap analysis, with identical machinery and the same global minimum margin  $6.009249 \times 10^{-5}$ ). The formulation of Section 3 requires avoidance of the  $k$ -space cut  $[0, 64]$  (Remark 2.3) on a path extended to the point  $i$  (where  $|s_2| > 261 > 64$  elementarily, Lemma 3.3); this strengthened certification is carried out by `cert2_path_k064.py` and yields the block counts in the table above.

### 6.3 Exactness certificates

The exactness of  $s_2(\tau_0) = 1$  and of  $s_2(\tau') = (47 + 45\sqrt{-7})/2$  is certified by `cert0_s2_eq_1.py` and `cert0_s2_n2pair.py`:

- the formal product identity (6) is re-checked as an exact integer  $q$ -series identity to order  $q^{37}$  (rational arithmetic);
- the sextic factorizations (25) and (28) are verified by exact polynomial division over  $\mathbb{Q}$ ;
- $j(2\tau_0)$  and  $j(2\tau')$  are enclosed by intervals with rigorous tails:  $|j(2\tau_0) + 3375| < 10^{-66}$  and  $|j(2\tau') + 3375| \leq 1.8 \times 10^{-91}$ , pinning the integers;
- the  $\lambda$ -values are locked by interval evaluation against the root separation:  $|\lambda(2\tau_0) - (1 + 3\sqrt{-7})/2| < 10^{-58}$  with separation 3.75, and  $|\lambda(2\tau') - (31 + 3\sqrt{-7})/32| \leq 1.2 \times 10^{-62}$  with separation  $3\sqrt{7}/16 = 0.4960\dots$

## 6.4 Script inventory

| script                           | content                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------|
| <code>cert0_s2.eq.1.py</code>    | exactness $s_2(\tau_0) = 1$ (series check, sextic factorization, $j$ and $\lambda$ locks)     |
| <code>cert2_path_k064.py</code>  | path certification for Theorem A (avoided set $[0, 64]$ ; self-test 20/20)                    |
| <code>cert2_path.py</code>       | companion: shorter path, avoided set $[-8, 8]$                                                |
| <code>verify_P1.py</code>        | (P1) at $\tau_0$ : theta identity to $q^{60}$ (exact integers), $T_1, T_4$ by Poisson rows    |
| <code>lvalue_g7.py</code>        | $L'(g_7, 0)$ to 50 digits; functional-equation self-check to 52 digits                        |
| <code>mahler_m.py</code>         | true $m(f+1)$ to 40 digits by direct torus integration                                        |
| <code>find_tau0.py</code>        | $\tau_0, s_2(\tau_0), \text{EK}(\tau_0)$ to 60 digits                                         |
| <code>samart_ek.py</code>        | EK formula; three-way validation at $k = 64$                                                  |
| <code>cert0_s2_n2pair.py</code>  | exactness $s_2(\tau') = (47 + 45\sqrt{-7})/2$ and the conjugate (100 dps locks)               |
| <code>verify_P1_n2pair.py</code> | (P1) at $\tau_w$ : 43 checks, all pass (60 dps, worst $ \text{diff}  = 1.2 \times 10^{-58}$ ) |
| <code>verify_n2_pair.py</code>   | three-way numerical confirmation of Theorem B (50 dps)                                        |
| <code>diag_ek.branch.py</code>   | wrong-sheet diagnostic of Remark 5.12                                                         |

All computations use `mpmath` interval arithmetic at 40–100 decimal digits. The certification scripts (`cert0_*`, `cert2_*`, `verify_P1*`) print PASS/FAIL per check and terminate with an all-checks-passed line; the remaining scripts print numerical values used for cross-checks.

**Code and certificate availability.** All scripts listed above are available in the public repository <https://github.com/huiminZheng-collab/samart-mahler>, archived at <https://doi.org/10.5281/zenodo.21711885>, and from the author upon request. The complete machine-readable certificate — every block enclosure, margin, and cap constant — is printed by `cert2_path_k064.py` and archived in the repository under `cert/output/`; re-running the scripts regenerates it from scratch, so no part of it is embedded only in this text.

## 7 Numerical confirmations

### 7.1 Theorem A

The three quantities

$$2m(f+1) \text{ (direct torus integration),} \quad \text{EK}(\tau_0) \text{ (Poisson-summed series),}$$

and  $8L'(g_7, 0)$  (from the functional equation) agree pairwise to at least 41 digits (`mahler_m.py`, `find_tau0.py`, `lvalue_g7.py`):

$$\begin{aligned} m(f+1) &= 0.4106864311156080448418263795931716559956\dots, \\ 4L'(g_7, 0) &= 0.4106864311156080448418263795931716559956\dots \quad (\text{difference } 6 \times 10^{-42}), \\ \text{EK}(\tau_0) &= 8L'(g_7, 0) \\ &= 0.8213728622312160896836527591863433119911586\dots \end{aligned}$$

As a control, the axis identity at  $k = 64$  was validated three ways (via EK, via direct integration, and via  $8L'(\eta(4\tau)^6, 0)$ ) to 50 digits (`samart_ek.py`).

## 7.2 Theorem B

The three quantities

$$n_2(k_+) \text{ (direct torus integration),} \quad \text{EK}(\tau_w), \quad \frac{4}{7}(54M_7 + d_7)$$

agree to  $\sim 10^{-50}$  (`verify_n2_pair.py`, `verify_P1_n2pair.py`):

$$n_2\left(\frac{47 \pm 45\sqrt{-7}}{2}\right) = \frac{4}{7}(54M_7 + d_7) = 4.13826815832141945836337668037433080930698376\dots$$

The Rogers hypergeometric form (3) of  $n_2$  (convergent at  $|64/k_{\pm}| = 1$ ) independently gives the same value to  $10^{-60}$ . The wrong-sheet value of Remark 5.12,  $\text{EK}(\tau') = (8/7)(44M_7 - d_7) = 3.22268\dots$ , differs from  $n_2(k_+)$  by  $0.91558\dots$ .

## 7.3 Reference values

$$\begin{aligned} L'(g_7, 0) &= 0.10267160777890201121045659489829291399889482708922\dots \\ L(g_7, 3) &= 0.6875636561025306425038995842156425788871541\dots \\ d_7 = L'(\chi_{-7}, -1) &= 1.69770245700177544677125306614726156034690092\dots \\ L(\chi_{-7}, 2) &= 1.1519254705444910471\dots \\ \zeta_K(2) &= 1.8948414489688065289713480740538331491752358\dots \end{aligned}$$

## A Exact polynomials and expansions

The sextic of §4.2 and §5.2:

$$\Phi(X) = 256(X^2 - X + 1)^3 + 3375X^2(1 - X)^2,$$

i.e.

$$\Phi(X) = 256X^6 - 768X^5 + 4911X^4 - 8542X^3 + 4911X^2 - 768X + 256$$

(note the palindromic coefficients) factors over  $\mathbb{Q}$  as

$$\Phi(X) = (X^2 - X + 16)(256X^4 - 512X^3 + 303X^2 - 47X + 16),$$

and splits completely over  $\mathbb{Q}(\sqrt{-7})$  into three quadratic factors:

$$\Phi(X) = (X^2 - X + 16)(16X^2 - X + 1)(16X^2 - 31X + 16),$$

with roots

$$\frac{1 \pm 3\sqrt{-7}}{2} \in X^2 - X + 16, \quad \frac{1 \pm 3\sqrt{-7}}{32} \in 16X^2 - X + 1, \quad \frac{31 \pm 3\sqrt{-7}}{32} \in 16X^2 - 31X + 16.$$

The roots of the last two factors are approximately

$$0.96875 \pm 0.2480391854123053678595265 i, \quad 0.03125 \pm 0.2480391854123053678595265 i$$

(80-digit **polyroots**). The two locks used in the text:

- $\lambda_0 = (1 + 3\sqrt{-7})/2 = 0.5 + 3.968626966596885885752424 i$  (Theorem A): minimum distance 3.75 to the other roots of  $\Phi$ ;
- $\lambda_1 = (31 + 3\sqrt{-7})/32$  (Theorem B): minimum distance  $3\sqrt{7}/16 = 0.4960784 \dots$  to the other roots, attained at its conjugate  $(31 - 3\sqrt{-7})/32$ .

Expansion of  $s_2$  (Lemma 2.4):

$$s_2 = q^{-1} + 24 + 276q + 2048q^2 + 11202q^3 + 49152q^4 + 184024q^5 + 614400q^6 + 1881471q^7 + \dots$$

Expansion of  $\lambda(2\tau)$ :

$$\lambda(2\tau) = 16q - 128q^2 + 704q^3 - 3072q^4 + 11488q^5 - 38400q^6 + 117632q^7 - 335872q^8 + \dots$$

## B The theta identity (8)

Both sides of (8) lie in  $S_3(\Gamma_0(7), \chi_{-7})$ : the left side is the eta product  $\eta(\tau)^3 \eta(7\tau)^3$  (standard eta-product criteria); the right side is the theta series of the Hecke Größencharakter  $\psi((\alpha)) := \alpha^2$  of  $K = \mathbb{Q}(\sqrt{-7})$  (conductor (1), type (2, 0)), which by Hecke's theorem is a cusp form of weight 3, level  $|\text{disc } K| = 7$  and nebentypus  $\chi_{-7}$ .

The Sturm bound for weight 3 and level 7 is  $\lfloor 3 \cdot [\text{SL}_2(\mathbb{Z}) : \Gamma_0(7)] / 12 \rfloor = \lfloor 3 \cdot 8 / 12 \rfloor = 2$ , so equality of the coefficients of  $q^1$  and  $q^2$  already proves (8). `verify_P1.py` checks the first 60 coefficients *exactly* as integers: the coefficient of  $q^N$  on the right is

$$\frac{1}{2} \sum_{\substack{m, n \in \mathbb{Z} \\ m^2 + mn + 2n^2 = N}} ((m^2 - 2n^2) + (2mn + n^2)\omega),$$

whose  $\omega$ -component vanishes for every  $N$  (pairing  $\alpha$  with its conjugate) and whose rational component reproduces (7).

Alternatively,  $\dim S_3(\Gamma_0(7), \chi_{-7}) = 1$  (there are no oldforms since  $S_3(\text{SL}_2(\mathbb{Z})) = 0$ , and the single newform orbit is 7.3.b.a [14]), and both sides are normalized ( $a_1 = 1$ : from the Jacobi expansion on the left, and from  $\alpha = 1$  contributing  $\frac{1}{2} \cdot 1^2 q^1$  on the right).

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